

Customize Ecommerce Website

(Fit Forge Apparel)



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Final Year Project Report

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A Customize E-Commerce Platform

Executive Summary

Fit Forge Apparel is e-commerce platform that incorporates real-time 3D personalization to completely transform the clothes shopping experience. This project proposal describes the goals, scope, and relevance of developing an interactive online store where customers may modify products like as sweatshirts and hosiery with cutting-edge technology like Three.js and React.js.

The Software Requirements Specification (SRS) specifies both functional and non-functional requirements, focusing on user authentication, product customization, order administration, and secure payment processing via the Payoneer API. The Software Design Specification (SDS) describes the system architecture, database design, and technological stack, which uses Laravel for backend development, MySQL for database management, and Firebase for media storage.

The prototype proves the system's viability, with a dynamic user interface, real-time 3D model rendering, and a simple purchasing experience. Fit Forge Apparel combines modern UI/UX principles, secure transaction protocols, and an efficient backend infrastructure to bridge the gap between traditional e-commerce and interactive shopping.

Positioned as a competitive solution in the personalized sportswear sector, the platform provides a seamless and engaging shopping experience while maintaining high levels of security, performance, and customer satisfaction.

FYDP Overview

FYDP Title: FIT FORGE APPAREL

Sr. No	Roll Numbers	Name	Signatures
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Table 1 Project Proposal Summary

FYDP Goals
The primary goal of Fit Forge Apparel is to develop a scalable and interactive e-commerce platform that specializes in customized sportswear. The project aims to provide real-time 3D customization to enhance user engagement and purchasing confidence. It will integrate secure payment processing, order tracking, and an intuitive shopping experience to align with modern e-commerce trends.
FYDP Objectives
Fit Forge Apparel will implement real-time 3D customization using React.js and Three.js, enabling interactive product design. Built on a secure, scalable backend with Laravel, the platform ensures robust performance. It features secure user authentication with role-based access, multiple payment gateways (including Payoneer & VisaAPI), and an efficient order tracking system with local couriers. The mobile-responsive UI/UX, powered by Tailwind CSS, delivers a seamless experience, while MySQL ensures data security and privacy. These features make Fit Forge Apparel a modern, secure, and user-friendly e-commerce solution.
FYDP Success Criteria
Fit Forge Apparel ensures seamless user authentication with secure registration, login, and role-based access. It offers efficient 3D customization, enabling real-time product personalization without delays. The platform provides reliable order processing and tracking, ensuring accurate order management. Built for optimized performance and security, it is fast, scalable, and future-ready. Additionally, the user-friendly interface is intuitive and responsive, delivering a smooth experience across all devices.
Assumptions:
• Users are familiar with basic e-commerce operations and customization tools. • A stable internet connection is available for optimal user experience. • Customers trust online payment methods, including Payoneer and other integrated payment gateways. • Local courier services will support accurate and real-time order tracking.
Risks & Obstacles
Fit Forge Apparel faces technical complexity in implementing real-time 3D customization, requiring extensive optimizations. Security threats, such as data breaches, unauthorized access, and payment fraud, demand robust protective measures. Transaction failures due to payment gateway integration issues could disrupt seamless transactions. Logistics and delivery challenges, including delays and tracking inaccuracies, may impact customer satisfaction. Additionally,

performance bottlenecks must be addressed to ensure the platform remains fast and scalable under heavy traffic conditions.

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Date: October 07, 2024

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Chapter No 1

Project Proposal

1.1. Introduction

Fit Forge Apparel is a 3D customization e-commerce website that allows customers to design and personalize hoodies and active wear in real time before purchasing. It integrates React (Three.js) for the frontend, Laravel for the backend, and Blender for 3D modeling to provide an interactive shopping experience. Traditional e-commerce platforms lack real-time personalization, which leads to customer dissatisfaction. Fit Forge Apparel addresses this by allowing customers to change the colors, textures, logos, and sizes on high-fidelity 3D models. The platform provides a smooth, fast, and responsive UI, secure transactions, and cross-device compatibility. Fit Forge Apparel's innovative strategy increases consumer engagement, minimizes return rates, and transforms the online Apparel purchasing experience.

1.2. Background

In today's e-commerce world, personalization is an important driver of customer pleasure and competitive advantage. Traditional online platforms sometimes use static graphics and limited customization choices, leaving customers unsure about how a customized product will actually look. Advancements in 3D rendering technology have created new possibilities for interactive, real-time customization. Fit Forge Apparel takes advantage of these developments to create a dynamic platform where users can create and customize hoodies and active wear in a realistic 3D environment. This project uses new web technologies to provide an immersive experience, solving deficiencies in traditional e-commerce systems and attempting to reduce product returns while increasing user engagement.

1.3. Problem Statement

Traditional online apparel platforms depend on static images and limited customization choices, providing clients an inadequate picture of how customized products will look. This often results in high return rates, when consumers receive things that do not fulfill their expectations, causing major customer dissatisfaction. Customers are unwilling to spend in customized apparel without a realistic, interactive preview, which reduces conversion rates and decreases trust in the online buying experience overall.

Fit Forge Apparel fills this essential gap by providing a real-time 3D customization platform that allows customers to design hoodies and active wear while receiving instant visual feedback. The platform seeks to provide a smooth, immersive experience by integrating modern technologies such as React, Three.js, and Draco compression, which not only increases user engagement but also lowers product return rates. The addition of secure payment processing through Payoneer & Visa API, as well as features such as order tracking and a review system, provides a comprehensive solution that fulfills both customer and business needs in today's competitive e-commerce website.

1.4. Stakeholders & Interests

Table 2 Stakeholders & Interests

Customer Type	Description & Interests	Potential Market Size
Customers (End Users)	Individuals looking for personalized apparel. They want an easy 3D customization experience that allows them to change product colors, sizes, and textures, providing a realistic preview before purchasing. This improves confidence and lowers returns.	Over 100 million potential customers globally
Small & Medium-Sized Apparel Brands	Emerging apparel firms and independent designers seeking a low-cost platform to provide personalized products without a significant investment in technology. They require an easy-to-use solution that enables them to differentiate their products in a competitive marketplace.	Approximately 500,000 to 1 million potential brands and designers worldwide
E-Commerce Businesses	Established online businesses want to increase client interaction with interactive 3D customisation options. They want to increase conversions and reduce product returns by providing a more engaging shopping experience.	Estimated 5,000+ online retail platforms globally
Sports Teams & Organizations	Organizations and sports teams that need custom-branded clothes for uniforms and promotional products. They value high-quality, scalable systems that can manage large orders while offering consistent branding options.	Tens of thousands of potential corporate and organizational clients globally.

1.5. Objectives

The primary objective of Fit Forge Apparel is to develop a comprehensive e-commerce platform that enhances customer experience through real-time 3D product customization and efficient order management. The platform aims to bridge the gap between user expectations and actual product delivery.

Key Objectives:

- **Interactive 3D Customization:**
Develop a real-time 3D customization tool with Three.js with Draco compression that allows users to create hoodies and active wear by changing colors, sizes, and textures while receiving high-fidelity visual feedback.
- **Enhanced User Experience:**
Create an intuitive and responsive UI with React and Tailwind CSS that streamlines the customization process, minimizes product returns, and provides helpful tooltips and visual guides for first-time users.

- **Robust and Secure Backend:**
With React and Tailwind CSS, you can create a simple and responsive UI that simplifies the customization process, reduces product returns, and provides helpful tooltips and visual guidance for first-time users.
- **Secure Payment Processing:**
Integrate Payoneer & Visa API with other payment gateways to ensure PCI-DSS compliance, safe transactions, and an easy checkout process.
- **Scalable Architecture and Continuous Improvement:**
Design a scalable system that can handle heavy traffic and potential expansion into new product categories, as well as continuous integration, automated testing, regular backups, and active user input to ensure for a long time reliability and performance.

1.6. Scope

The scope of Fit Forge Apparel encompasses the design, development, and deployment of an interactive e-commerce platform with real-time 3D product customization, efficiently. The system will provide a user-friendly and visually immersive shopping experience, ensuring better engagement and reduced return rates. The project scope includes:

- **Real-Time Customization:**
Customers can change product colors, sizes, textures, and designs in real time using an interactive 3D tool developed with React and Three.js.
- **Secure User Interactions:**
Using secure user registration, login, and JWT-based authentication to assure authorized access.
- **Order Processing & Tracking:**
Enables secure payment processing using integrated gateways like Payoneer & Visa API, as well as real-time order tracking.
- **Review System:**
Users can write reviews on products, increasing trust and feedback.
- **Responsive Design:**
Ensures the system is accessible and fully responsive on PCs, tablets, and smartphones.
- **Admin Dashboard:**
Providing administrators with an admin dashboard for managing users, products, and orders.

1.6.1 Limitations

- **Product Range:**
The platform is currently limited to customizing and selling only hoodies and active wear.
- **Feature Scope:**
Advanced features like subscription models, AI-driven design suggestions, and AR/VR functionalities are not included.
- **Device Compatibility:**

Although optimized for modern browsers and devices, performance might be reduced on older hardware or networks.

- **Payment Methods:**

Payment integration is limited to major gateways (Payoneer & Visa API) and may not cover all regional payment options.

1.7. Assumptions

The development and implementation of Fit Forge Apparel rely on critical assumptions to ensure the system aligns with business goals and user requirements, providing a seamless and reliable e-commerce platform:

- **Device and Connectivity:**

Users will have access to modern devices (e.g., PCs, tablets, smartphones) with stable internet connectivity, enabling smooth interaction with real-time 3D customization features.

- **User Familiarity:**

Customers are assumed to have basic familiarity with e-commerce processes like browsing, customization, and checkout, reducing the learning curve and enhancing engagement.

- **Third-Party Services:**

External APIs (e.g., Payoneer, Visa API) will remain available, functional, and well-documented, ensuring seamless integration for payment processing and other functionalities.

- **Technical Expertise:**

The development team possesses proficient expertise in React, Three.js, Laravel, and Blender, ensuring a high-performance, scalable system.

- **Market Trends:**

The market will continue to favor personalized, interactive shopping experiences, aligning with the platform's focus on real-time 3D customization.

1.8. Risks

The development and deployment of Fit Forge Apparel come with several potential risks that could impact the functionality, security, and user experience of the platform. These risks must be identified and mitigated to ensure a smooth and reliable e-commerce system.

- **Performance Challenges:**

High-resolution 3D models may slow down the system; mitigated by employing Draco compression and optimization techniques.

- **Security Vulnerabilities:**

Potential risks like SQL injection and XSS attacks; mitigated by implementing secure coding practices and regular security audits.

- **Scalability Issues:**

High traffic might strain system resources; mitigated by using cloud hosting with autoscaling capabilities.

- **Integration Risks:**

Reliance on external APIs (e.g., Payoneer & Visa API) for payment processing may lead to integration challenges; mitigated by thorough testing and contingency planning.

- **User Adoption:**
Users may face difficulties with the new interface; mitigated by providing comprehensive onboarding tutorials and user support.

1.9. Success Criteria

The success of the Fit Forge Apparel system is determined by the implementation and effectiveness of the following key features. These features ensure system functionality, user experience, performance, and security, making the system acceptable for deployment.

Table 3 Project Success Criteria

User Registration and Authentication: Secure login and sign-up functionality. Core System Functionalities: Product listing, customization, cart management, and order processing. Data Management: Efficient handling of products, users, orders, and payments. Customization Feature: Interactive 3D customization of sportswear. Search and Filter Options: Users can find products easily using categories and filters. Speed and Responsiveness: Optimized performance with fast page loading times. Scalability: System must support increasing users, products, and orders. Data Privacy: Secure handling of user and payment data. Access Control: Role-based access for customers, admins, and suppliers. User Interface Design: Interactive, visually appealing, and intuitive UI. Cross-Platform Compatibility: Functionality across desktops, tablets, and smartphones. Error Handling: Proper notifications and recovery mechanisms for failed operations. Third-Party Integration:

Payment gateways, shipping services, and authentication providers.

Deployment Readiness:

Fully functional, tested, and secure for public use.

1.10. Tools, Libraries & Technology

Table 4 Tools Technologies and Libraries

Category	Name	Version	Rationale
Tools	VS Code	Latest	Primary code editor for development.
	Blender	Latest	3D model optimization and rendering.
	Postman	Latest	API testing and debugging.
	Git & GitHub	Latest	Version control and collaboration.
Libraries	React.js	Latest	Frontend framework for UI development.
	Three.js	Latest	3D rendering for product customization.
	Tailwind CSS	v4	Utility-first styling for responsive UI.
	Laravel	Latest	Backend framework for API development.
	MySQL	Latest	Database for storing user, product, and order data.
Payment	Payoneer & Visa API	Latest	Payment gateway for transactions.

1.11. Work Division

Table 5 Project Team Members Work Division

Sr. No	Roll Number	Name	Role Assignment & Work Division
1.	069647/2108	Bilal Ahmed	Frontend, Backend
2.	069640/2103	Ali Haider	Frontend, Backend
3.	069569	Shehroze Waseem	Documentation & User Interface

1.12. Conclusion

Fit Forge Apparel revolutionizes e-commerce with real-time 3D customization, enabling users to personalize apparel before purchase. The project enhances customer confidence, reduces return rates, and delivers a unique, interactive shopping experience. Through clear objectives like seamless customization, secure transactions, and scalable infrastructure, the platform ensures efficiency for users and administrators. Assumptions like user familiarity with customization tools and stable internet connectivity guide execution, while risks like performance bottlenecks and security vulnerabilities are mitigated. By achieving success criteria, efficient order processing, responsive UI, high data security, and third-party integrations, Fit Forge Apparel emerges as an innovative, user-centric platform.

Chapter No 2

Literature Review

2.1 Literature Survey

Survey on the Evolution and Challenges of 3D Customization in Activewear are as followed:

2.1.1 Evolution of 3D Customization in Activewear

Over the years, 3D customization in the activewear industry has evolved significantly, from basic 2D previews to fully interactive, real-time 3D configurators. The key developments include:

- **Early Customization (Before 2010):**
Simple 2D mockups with limited color and text options.
- **Intermediate Phase (2010-2020):**
The introduction of 3D previews, allowing users to rotate and inspect designs.
- **Current Trends (2020-Present):**
Real-time WebGL-based 3D customization (Three.js, Babylon.js) with dynamic textures, live rendering, and even AR/VR integration for enhanced visualization.

Companies like Nike by You, virtualthread.io, iTailor[1], owayo, icmfactory, Spized, and Kit Builder have contributed to this evolution by improving user experience, real-time design accuracy, and material simulation.

2.1.2 Key Elements of a Successful 3D Customization Website

According to market leaders such as Nike by You, virtualthread.io, iTailor, owayo, icmfactory, Spized, and Kit Builder, the main aspects required for a successful 3D activewear customization platform are:

- **Real-Time 3D Rendering:**
Interactive 3D models that respond to user customization (colors, logos, patterns).
- **Seamless UI/UX:**
A user-friendly, responsive, and smooth experience that guides customers effortlessly.
- **Material & Texture Simulation:**
Accurately displaying how fabrics stretch, reflect light, and adapt to designs.
- **Integration with E-commerce:**
A streamlined checkout process with direct-to-manufacturing capabilities.
- **Multi-Platform Compatibility:**
Web and mobile accessibility with minimal performance lag.

2.1.3 Challenges Faced by 3D Customization Platforms

Despite its benefits, firms (like Fit Forge Apparel) may face the following problems when deploying a 3D customization platform:

- **Technical Limitations & Performance:**
 - Real-time 3D rendering requires high computational power, causing slow load times for complex designs.
 - Balancing quality and speed is crucial to avoid losing customers.
- **Development Complexity & Cost:**
 - Implementing Three.js, WebGL for smooth 3D customization is challenging.

- High investment in backend infrastructure (Laravel).
- **User Experience & Engagement:**
 - Customers unfamiliar with 3D tools might find the interface confusing.
 - Requires an intuitive UI/UX, similar to how Nike By You and VirtualThread.io simplifies its customization process.
- **Accurate Fabric Simulation:**
 - Real-world materials behave differently from digital previews.
 - Developing realistic textures and draping physics (e.g., using Blender or Cloth Physics APIs) is a challenge.
- **Manufacturing Integration:**
 - Ensuring that customized designs translate perfectly into production without errors.
 - Automated order-to-manufacture processes must minimize defects and return rates.
- **Market Competition & Adoption:**
 - Competing with established players like VirtualThread.io, iTailor, Kit Builder requires differentiation.

2.2 Related Work:

The development of 3D personalization in activewear has transformed e-commerce by allowing customers to personalize sportswear in real time. This section examines the existing platforms and technologies that have influenced the growth of 3D customization in sportswear.

2.2.1 Existing Platforms for 3D Customization

Several companies have successfully integrated **3D configurators** into their online stores, offering users the ability to customize and preview products before purchase. Notable platforms include:

1. **Nike By You:**
Allows users to personalize shoes with custom colors, materials, and engravings. Its realistic 3D previews enhance user engagement and confidence. [1]
2. **Owayo:**
Specializes in custom team jerseys for sports like cycling, soccer, and esports. Its 3D Kit Designer provides a 360-degree interactive preview for colors, patterns, and logos.[6]
3. **ICM Factory Direct:**
Offers a real-time 3D configurator for sportswear, enabling users to import logos, select colors, and add text. Focused on bulk orders and production efficiency.[3]
4. **VirtualThread.io:**
Uses AI-powered 3D garment visualization to create realistic prototypes, reducing manufacturing waste and improving design accuracy.[4]
5. **iTailor:**
A leader in custom tailoring, offering real-time 3D customization for shirts, suits, and activewear. Focuses on fabric selection and precise measurements.
6. **Spized:**
A European platform for sportswear customization, featuring 3D previews for jerseys, hoodies, and fitness apparel. Optimized for team orders and bulk production.[2]
7. **Kit Builder:**

Provides a 3D clothing design tool with logo placement, text customization, and realistic fabric textures. Enhances the shopping experience with 360-degree views.[5]

2.2.2 Limitations of Existing 3D Customization Platforms

Despite advancements, current platforms face several challenges:

- **Performance & Loading Speed:**
 - High-quality 3D models and textures slow down websites, increasing drop-off rates.
 - Web-based configurators demand significant computational power, limiting accessibility on low-end devices.
- **User Experience Challenges:**
 - Complex and unintuitive tools overwhelm users, leading to low engagement.
 - Lack of tutorials or onboarding guides further discourages first-time users.
- **Fabric & Material Simulation Issues:**
 - Digital previews often fail to match real-world fabrics, causing discrepancies between designs and final products.
 - Accurately displaying fabric stretch, draping, and textures remains a technical challenge.
- **Manufacturing & Order Integration:**
 - Poor integration with production facilities leads to errors in printing, stitching, or material selection.
 - Inefficient translation of 3D designs into production-ready formats increases processing times.
- **High Competition:**
 - Industry leaders like Nike By You and Owayo set high standards, making it difficult for new platforms to gain market share.

2.2.3 Proposed Solution

Fit Forge Apparel addresses these limitations through innovative solutions:

- **Optimized 3D Rendering:**

Using Three.js and compressed assets for fast, smooth interactions.
- **Beginner-Friendly UI:**

Simplifying customization with an intuitive interface for first-time users.
- **Realistic Fabric Simulation:**

Enhancing texture accuracy with advanced shading, cloth physics, and rendering techniques.
- **Seamless Backend Integration:**

Leveraging Laravel for real-time order tracking, updates, and automated production workflows.

- **Market Differentiation:**

Reducing design time and improving personalization to stand out in a competitive market.

2.3 Gap Analysis:

Gap analysis is required to determine the differences between current 3D customization platforms and the proposed Fit Forge Apparel system. This section discusses the limits of current solutions and how Fit Forge Apparel will overcome them through creative techniques.

2.3.1 Identified Gaps in Existing 3D Customization Platforms

Despite advancements in 3D customization, current platforms face several shortcomings that affect performance, user experience, and production efficiency. The major gaps include:

- **Performance & Loading Speed Issues**
 - Platforms like Nike By You and Owayo offer high-quality 3D rendering but suffer from slow load times, especially on low-end devices.
 - WebGL-based configurators consume significant computational power, causing lag and crashes on mobile devices.
- **Limited User Experience & Accessibility**
 - Many platforms feature complex interfaces that require technical expertise, discouraging first-time users.
 - Lack of guided customization workflows leads to user confusion and mid-process abandonment.
 - Some platforms (e.g., iTailor, ICM Factory) lack mobile responsiveness, limiting accessibility.
- **Fabric Realism & Texture Simulation Issues**
 - Existing 3D configurators struggle to replicate real-world fabric textures and stretch effects accurately.
 - Discrepancies between digital previews and final products result in customer dissatisfaction and higher return rates.
- **Weak Integration with Manufacturing & Order Processing**
 - Many platforms lack efficient workflows for converting 3D designs into production-ready formats.
 - While Nike By You and VirtualThread.io excel in supply chain integration, others struggle with printing and stitching accuracy.
 - Real-time tracking for customized orders is often missing.

2.3.2 Proposed Solutions to Bridge the Gaps

To address these challenges, Fit Forge Apparel will introduce the following innovations:

Table 6 Proposed Solution to Bridge the Gaps

Identified Gaps	Fit Forge Apparel Solutions
Slow 3D rendering and high loading times	• Three.js optimization & texture compression for faster load times. • Lazy loading for assets to improve mobile performance.
Complex UI leading to user frustration	• Simplified step-by-step UI. • AI-assisted customization suggestions. • Mobile-friendly design for accessibility.
Fabric texture and material realism issues	• Advanced physics-based cloth simulation. • Dynamic lighting and shading for accurate fabric previews.
Lack of AI-powered customization	• AI-based design recommendations to speed up customization. • Real-time previews based on popular trends and user preferences.
Weak integration with manufacturing	• Automated order processing system. • Direct manufacturer integration for print accuracy. • Real-time tracking system for orders.

2.4 Summary

The development of 3D customization in activewear has altered e-commerce by allowing for real-time visualization, customizable designs, and interactive user participation. Nike By You, Owayo, and VirtualThread.io use WebGL, Three.js, and AI techniques to provide 360° previews and modification of colors, logos, and textures. However, issues like as slow loading times, confusing interfaces, inaccurate fabric rendering, and inadequate production integration persist, causing customer dissatisfaction and differences between digital previews and finished items. Fit Forge Apparel fills these gaps by introducing high-performance 3D visualization, AI-driven design advice, realistic fabric simulation, and streamlined manufacturing processes. These advances provide a speedier, more intuitive, and highly personalized customization experience, lowering return rates and increasing consumer satisfaction in the sportswear market.

Chapter No 3

Software Requirements Specification

3.1 Requirements Analysis

The Requirements Analysis section is a structured breakdown of the system requirements for the 3D Customization E-Commerce Platform. This document describes the functional and non-functional criteria needed to create a seamless and dynamic 3D product customizing experience. The study ensures that all stakeholders, including customers, and administrators, are fully aware of the system's capabilities. Functional needs include user identification, product customization, order processing, review system, order tracking, and Payoneer & Visa API payment integration, while non-functional requirements include performance, security, and usability. This section goes into greater detail about external interfaces, system constraints, and use case modeling, providing a complete specification for the platform's development and implementation. The goal is to establish a clear and structured foundation that will assist the development team in creating an efficient, scalable, and user-friendly system that meets both business and user needs.

3.2 User classes and characteristics

Identification of the various user classes that will use this product and describe their pertinent.

Table 7 User classes & Characteristics

User Class	User Characteristics
Administrator	Manages user accounts, checks product listings, keeps the website working, ensures security, and controls order tracking and payment processes.
Customers	Browses the website, explores available 3D customization options, views reviews, but cannot make purchases without registration.

3.3 Requirement Identifying Technique

Fit Forge Apparel is an interactive 3D Customization E-Commerce Website. Use Case Modeling is an effective technique for identifying functional requirements. This technique ensures that all interactions between users and the system are clearly defined, making it easier to understand functionalities and workflows.

3.4 Functional Requirements

This section outlines the functional requirements of the Fit Forge Apparel system in a structured format. Each feature is described along with its corresponding functional requirements, ensuring clarity in system behavior and expected functionalities.

Table 8 Function Requirements Specification for Product Customization

Identifier	FR-1
Title	Product Customization
Requirement	The buyer shall be able to customize hoodies and active wear in real time using interactive 3D models, modifying colors, sizes, and textures with immediate visual feedback.
Source	User Needs
Rationale	Enhances user experience by providing an immersive, accurate preview, reducing return rates.
Business Rule (if required)	Only predefined customization options are available.
Dependencies	FR-7 (Performance Optimization)
Priority	High

Table 9 Function Requirements Specification for User Management

Identifier	FR-2
Title	User Management
Requirement	The user shall be able to create an account, log in securely, and manage their profile.
Source	Project Scope
Rationale	Ensures secure access and efficient management of user data.
Business Rule (if required)	Passwords must meet security criteria; email verification is required.
Dependencies	-
Priority	High

Table 10 Function Requirements Specification for Payment Processing

Identifier	FR-3
Title	Payment Processing
Requirement	The system shall securely process payments through an integrated gateway, including Payoneer, ensuring PCI-DSS compliance and a smooth transaction experience.
Source	Business Logic
Rationale	Provides a secure and efficient checkout experience, increasing customer trust.
Business Rule (if required)	Transactions must comply with PCI-DSS; use 2FA for Payoneer transactions.
Dependencies	FR-4
Priority	High

Table 11 Function Requirements Specification for Order Management

Identifier	FR-4
Title	Order Management
Requirement	Buyers shall be able to add customized products to the cart, place orders, and track order status in real time.
Source	User Needs
Rationale	Enhances transparency and streamlines the purchasing process.
Business Rule (if required)	Order status updates must be provided in real time.
Dependencies	FR-1,FR-3
Priority	High

Table 12 Function Requirements Specification for Inventory Management

Identifier	FR-5
Title	Inventory Management
Requirement	The system shall maintain an inventory of available products, updating stock levels automatically as orders are processed.
Source	Business Requirements
Rationale	Prevents overselling and ensures accurate tracking of product availability.
Business Rule (if required)	Inventory levels must update in real time.
Dependencies	FR-4
Priority	Medium

Table 13 Function Requirements Specification for Shipping & Delivery

Identifier	FR-6
Title	Shipping & Delivery
Requirement	The system shall provide shipping options and track delivery status for each order, enabling buyers to receive real-time updates on shipment progress.
Source	User Needs / Business Logic
Rationale	Enhances customer satisfaction by providing timely delivery information.
Business Rule (if required)	Shipping options must comply with regional regulations and carrier guidelines.
Dependencies	FR-4, FR-5
Priority	Medium

Table 14 Function Requirements Specification for Performance Optimization

Identifier	FR-7
Title	Performance Optimization
Requirement	The system shall optimize performance across all modules by ensuring fast page load times (<3 seconds), efficient 3D model rendering (<5 seconds) using Draco Compression, and a response time of less than 1 second for user interactions.
Source	Technical Requirement
Rationale	Ensures a seamless user experience, even under high load, and supports overall system efficiency.
Business Rule (if required)	All critical performance benchmarks must be met under normal conditions.
Dependencies	FR-1, FR-3, FR-4
Priority	High

3.5 Non-Functional Requirements

The Non-Functional Requirements section defines the system's performance, security, usability, and scalability expectations. Key aspects include high availability and reliability, secure authentication and data protection, fast response times for seamless user experience, scalability to handle increasing traffic and transactions, and compatibility across multiple devices and browsers. These quality attributes ensure the system operates efficiently and meets user expectations. Detailed non-functional requirements are documented in the following tables.

Table 15 Non Functional Requirements Specifications

Identifier	Requirement	Description	Priority
NFR-1	Performance	The system must support up to 500 concurrent users without significant performance loss. Non-3D pages must have a page load time of no more than two seconds. To maximize performance, 3D models must load in less than 3 seconds utilizing Draco Compression.	High
NFR-2	Scalability	The system must facilitate the addition of new product categories without requiring large code changes. The backend (Laravel) will use MVC architecture for ease of maintenance. The system	High

		must be able to scale to serve over 10,000 registered users.	
NFR-3	Security	The system has to use JWT authentication to protect user sessions. The system must prevent Cross-Site Scripting (XSS) and SQL Injection attacks.	High
NFR-4	Availability	The system shall be available 24/7, with downtime limited to 2 hours per month for maintenance. In case of system failure, an error message with estimated recovery time should be displayed.	High
NFR-5	Usability	Tailwind CSS will be used to design the interface, ensuring that it is responsive and user-friendly. Users should be able to easily customize products, submit feedback, and track orders (within 5 steps).	medium
NFR-6	Maintainability	The system shall follow a modular architecture to allow easy updates and maintenance.	High
NFR-7	Compatibility	The system must be responsive and functional on computers, tablets, and smartphones.	High

3.5.1 Reliability

Requirements for how frequently the software fails. The measurement is frequently represented in MTBF (mean time between failures), and Mean time to recover. Make sure to include the repercussions of software failure, how to prevent failure, a method for error detection, and a strategy for remedy.

Table 16 Reliability Specification

Identifier	Requirement	Description	Priority
NFR-9	Mean Time Between Failures (MTBF)	The system must have an MTBF of at least 500 hours to ensure minimal downtime and consistent functioning.	High
NFR-10	Mean Time to Recover (MTTR)	To minimize disturbance, the system must ensure that in the event of a breakdown, the recovery time does not exceed two hours.	High

NFR-11	Data Backup & Recovery	To avoid data loss, the system must execute incremental database backups every day and full backups once a week. In the event of a critical failure, the system must allow automatic rollback procedures that restore the previous stable state.	High
NFR-12	Error Detection Mechanism	The system must integrate real-time error logging and monitoring utilizing technologies such as Sentry or Log Rocket to spot issues early on.	High
NFR-13	Data Integrity	The system must ensure that 99.9% of transactions are handled correctly and without data corruption or loss.	High
NFR-14	Security & Crash Protection	To avoid crashes and illegal access, the system must have input validation, exception management, and security against SQL injection and XSS attacks.	High

3.5.2 Usability

The system should be simple to understand and operate, providing a consistent experience for users of various ability levels. Users should be able to personalize a product and place an order in 5 interactions. Error messages should be clear and actionable, bringing users to a resolution without annoyance. To effectively accommodate all users, the interface will adhere to accessibility requirements such as high-contrast modes, keyboard navigation, and screen reader compatibility.

Table 17 Usability Specification

Identifier	Requirement	Description	Priority
NFR-15	Ease of Learning	The system will give an interactive onboarding instruction to new users, guiding them through the 3D customizing process.	High
NFR-16	Ease of Use	The user interface should follow straightforward navigation patterns, allowing users to finish product customization in 5 interactions.	High
NFR-17	Error Avoidance	The system must feature real-time validation for user inputs to prevent invalid selections during the customizing process.	Medium

NFR-18	Efficiency of Interaction	The system shall ensure that a complete product customization and checkout process can be completed within 3 minutes under normal conditions.	High
NFR-19	Responsive Design	The website should be fully responsive, allowing for smooth use on PCs, tablets, and mobile devices.	High
NFR-20	Feedback Mechanism	The system shall provide instant feedback when users interact with customization options, submit reviews, track orders, or encounter payment issues through Payoneer & Visa API.	High

3.5.3 Performance

The system should provide quick and efficient performance throughout all operations. Product personalization should take less than three seconds, and order placement, including payment processing, should take no longer than ten seconds. The website should be able to manage at least 500 concurrent users without performance decrease. The server response time for product searches should be fewer than two seconds to ensure a pleasant and responsive user experience.

Table 18 Performance Specification

Identifier	Requirement	Description	Priority
NFR-21	Page Load Time	Under normal network conditions, the system should load the homepage in 3 seconds or less.	High
NFR-22	3D Model Rendering Time	The 3D customizing tool must load and display models in 5 seconds, with Draco compression for optimized performance.	High
NFR-23	Response Time for User Actions	The system must provide a response within 1 second for interactive actions like color selection, texture changes, and size adjustments in the 3D customization tool.	High
NFR-24	Concurrent Users Handling	The system shall support at least 5000 concurrent users without significant performance degradation.	High

NFR-25	Database Query Execution Time	Under normal load, all database queries must execute in less than two seconds.	High
NFR-26	Review Submission Response Time	To provide an engaging user experience, the system must process review submissions and update the review area in less than one second.	High
NFR-27	Order Tracking Update Time	When an order's status changes, the system must update the tracking information in real time (within 1 second).	High
NFR-28	Payment Processing (Payoneer & Visa API) Response Time	The system must process payments through the Payoneer & Visa API interface within 5 seconds, ensuring a smooth and safe transaction experience for buyers.	High

3.5.4 Security

By safeguarding user information, payment details, and order history, the system ensures data confidentiality, integrity, and availability. Unauthorized access should be prohibited using multi-layer security methods, and the system should be resistant to common cyber threats like SQL injection and cross-site scripting. The effort required to breach system security should be at least 100 hours when using advanced attack methods. Data transmission must be encrypted, and access to sensitive information should be limited based on responsibilities and permissions.

Table 19 Security Specification

Identifier	Requirement	Description	Priority
NFR-29	User Authentication	All users must provide secure authentication before accessing any customized features.	High
NFR-30	Data Encryption	All sensitive data, including user passwords and payment information, must be protected in travel and at rest with AES-256 encryption.	High
NFR-31	Access Control	The system will use role-based access control (RBAC) to prevent unauthorized access to admin and customers functions.	High

NFR-32	SQL Injection Prevention	All database queries must be parameterized to prevent SQL Injection attacks.	High
NFR-33	Order Tracking API Security	The system must secure order tracking API endpoints so that only authenticated buyers may access order status information, protecting data privacy and integrity.	High
NFR-34	Payment Gateway Security (Payoneer & Visa API)	The system must ensure that all payments made through Payoneer & Visa API are handled securely, in accordance with PCI-DSS requirements and using secure session management.	High

3.6 External Interface Requirements

This section contains information to guarantee that the system communicates effectively with users and other hardware or software parts. A complicated system with several subcomponents requires a separate interface or system architecture specification. Material from other publications could be incorporated into the interface documentation using references. For example, it could refer to a hardware device handbook that includes the fault codes that the device may communicate to the program.

Table 20 External Interface Requirements Specification

Identifier	Requirement	Description	Priority
NFR-35	User Interface (UI)	The system must provide an interactive and user-friendly UI for 3D customization, utilizing React and Tailwind CSS to ensure a consistent experience across every device.	High
NFR-36	API Integration	The system will use Laravel API routes to perform CRUD operations on products, users, and orders.	High
NFR-37	File Upload Interface	To ensure performance, the system will allow customers to upload 3D models in .glb format as well as photos in .jpg and .png formats, with size constraints.	High
NFR-38	External Payment Gateway	The system must have payment gateways (Payoneer & Visa API) for safe transactions and Google OAuth for authentication.	Medium
NFR-39	Hardware Compatibility	The system must be responsive and functional on computers, tablets, and smartphones. The 3D customization tool must run well on devices with at least 4GB RAM.	High

3.6.1 User Interfaces Requirements

The system's user interface must follow GUI design standards, which ensure consistency in navigation, icons, fonts, and interactive features. It should be snappy and support screen resolutions ranging from 1366x768 to 4K to ensure a consistent experience across all devices. Standard navigation components, such as Home, Back, and Help buttons, should appear on every screen, ensuring a consistent menu layout for ease of use. The UI must use modern and clean fonts like DMSans or VMvirtual, as well as a color scheme that is consistent with the brand identity and has enough contrast to be readable.

To accommodate disabled users, accessibility features such as screen reader support, high contrast mode, and keyboard shortcuts should be included. Additionally, to improve usability, the interface should include conventional keyboard shortcuts such as Ctrl+S for Save, Ctrl+Z for Undo, and Enter for Submit. Error messages and alarms should be clear and user-friendly, with instructions on how to remedy difficulties. To provide a pleasant user experience, notifications must adhere to the same design pattern across all interfaces.

Table 21 User Interfaces Requirements

Identifier	Requirement	Description	Priority
NFR-40	GUI Standards	The system will feature a modern UI design using Tailwind CSS with a consistent theme for buttons, fonts, and icons.	High
NFR-41	Font & Color Scheme	To enhance accessibility, the system must support sans-serif fonts (e.g., Poppins, Roboto) and a dark/light mode toggle.	High
NFR-42	Screen Resolution	The UI should be fully responsive and optimized for mobile, tablet, and desktop devices (minimum resolution: 1366x768).	High
NFR-43	Navigation & Layout	A fixed navigation bar must be present, providing easy access to Home, Customizer, Products, Orders, and Profile sections.	Medium
NFR-44	3D Customization UI	The system must feature a real-time 3D model viewer (powered by Three.js) with interactive color, size, and texture customization.	High

3.6.2 Software Interfaces

Table 22 Software Interface Requirements Specification

Component	Description	Purpose	Priority
Operating System	Windows / Linux / macOS	The system must be compatible with the major operating systems.	High
Frontend Framework	React.js	Used to create the interactive user interface and 3D customizing UI.	High
Styling Library	Tailwind CSS	Offers a responsive and adaptable UI design.	High
3D Rendering Library	Three.js	Allows for real-time 3D model rendering and customization.	High
Backend Framework	PHP Laravel	Manages business logic, authentication, and database operations.	High

Database	MySQL	Securely stores user, product, and order information.	High
3D Model Compression	Draco Compression	Optimizes 3D models for faster loading.	High
Payment Gateway API	Payoneer & Visa API	Orders are processed securely online.	High
Authentication	JWT (JSON Web Token)	Ensures safe login and session management.	High

3.6.3 Hardware Interfaces

Table 23 Hardware Interfaces Requirements Specification

Component	Supported Device	Purpose	Communication Protocols
User Device	PC, Laptop, Tablet, Smartphone	Provides access to the 3D customization platform	HTTP/HTTPS
Graphics Processing Unit (GPU)	NVIDIA, Integrated GPUs	Improves 3D model rendering and customizing performance.	OpenGL, WebGL
Server Hardware	Cloud-based	Hosts backend and database for smooth functioning.	TCP/IP, HTTPS
Display Monitor	Full HD (1920x1080) and above	Enables high-resolution 3D visualization	HDMI, DisplayPort
Internet Connection	Broadband, Fiber, 4G/5G, Wi-Fi	Allows internet access to the 3D customizing platform.	TCP/IP, HTTPS

3.6.4 Communications Interfaces

Table 24 Communication Interface Requirements Specification

Communication Component	Description	Protocol/Technology	Purpose	Priority
Web Browser	The system should be accessible using modern web browsers.	Chrome, Firefox, Edge, Safari	Allows users to interact with the 3D customization platform.	High
HTTP/HTTPS	Enables safe data exchange between client and server.	TCP/IP, SSL/TLS	Protects sensitive user data and transactions.	High
Payment Gateway API	Enables secure online payments.	Payoneer & Visa API	Ensures order payment processing runs smoothly.	High

Cloud Storage API	Stores and retrieves 3D models and user information.	AWS S3, Firebase Storage	Offers scalable storage for customized designs.	High
Admin Dashboard Communication	Allows administrators to manage orders, users, and products.	REST API	Offers real-time monitoring and control.	High

3.7 Use case Analysis

3.7.1 Use Case 1 : Login & Registration

- **Specification:**

This use case enables users to create an account and log in securely to access the platform. The system verifies user credentials, checks for duplicate accounts, and ensures security through email verification.

- **Key Features:**
 - User authentication and account creation
 - Email verification process
 - Error handling for incorrect login credentials

Table 25 Use Case Description for Login & Registration

UC Identifier	UC1: Login & Registration
Requirements Traceability	FR-1
Purpose	Allow users to create an account and log in securely.
Priority	High
Preconditions	The user must provide a valid email address.
Postconditions	The user account is created, and login is successful.
Actors	Customer
Extends	N/A
Includes	UC2: Email Verification
Main Success Scenario	1. User accesses the login/registration page. 2. Inputs details (email, password, etc.). 3. System verifies input and checks for existing accounts. 4. If valid, the system creates the account and logs the user in. 5. User is redirected to the dashboard.
Alternate Flows	1. If the email is already registered, the system prompts the user to log in. 2. If login credentials are incorrect, an error message is displayed.
Exceptions	1. Registration is unavailable during system downtime. 2. The email verification service fails.

3.7.2 Use Case 2 : 3D Customization

- **Specification:**

This use case allows users to interactively customize products in a 3D environment. Users can modify colors, textures, logos, and sizes before previewing their designs in real-time.

- **Key Features:**
 - Real-time 3D visualization of products
 - Customization with color, texture, and logos
 - Saving or adding customized designs to the cart

Table 26 Use Case Description for 3D Customization

UC Identifier	UC2: 3D Customization
Requirements Traceability	FR-2
Purpose	Enable customers to customize their activewear in 3D.
Priority	High
Preconditions	The user must be logged in.
Postconditions	The user successfully customizes and saves a design.
Actors	Customer
Extends	N/A
Includes	UC3: Texture & Color Selection
Main Success Scenario	1. User selects a product for customization. 2. Applies color, texture, logo, and size preferences. 3. Previews the design in real-time 3D. 4. Saves or adds the design to the cart.
Alternate Flows	1. If customization fails, the user is prompted to retry. 2. If input is invalid, an error message is displayed.
Exceptions	1. 3D model fails to load due to a network issue. 2. Poor internet connection affects rendering performance.

3.7.3 Use Case 3 : Scroll, Select & Purchase

- **Specification:**

Users can browse the product catalog, view product details, and select items to purchase. The system ensures an intuitive shopping experience by allowing users to add products to the cart and proceed to checkout.

- **Key Features:**
 - Seamless product browsing and selection
 - Dynamic cart management
 - Integrated checkout and payment processing

Table 27 Use Case Description for Scroll, Select & Purchase

UC Identifier	UC3: Scroll, Select & Purchase
Requirements Traceability	FR-3
Purpose	Allow users to browse, select, and purchase products.
Priority	High
Preconditions	The user must have access to the product catalog.

Postconditions	The selected product is added to the cart and successfully purchased.
Actors	Customer
Extends	N/A
Includes	UC4: Payment Processing
Main Success Scenario	1. User scrolls through the product catalog. 2. Selects an item and views details. 3. Adds the item to the cart. 4. Proceeds to checkout and completes the purchase.
Alternate Flows	1. If an item is out of stock, the system notifies the user. 2. The user can remove items from the cart before purchase.
Exceptions	1. Product details fail to load due to a network error. 2. The checkout page crashes, requiring a restart.

3.7.4 Use Case 4 : Order Placement & Tracking

- **Specification:**

Once an order is placed, the system processes payments, assigns the order to a courier, and provides real-time tracking updates. Users can monitor their order status and receive delivery notifications.

- **Key Features:**
 - Secure payment processing
 - Integration with courier services for tracking
 - Notifications for shipment updates

Table 28 Use Case Description for Order Placement & Tracking

UC Identifier	UC4: Order Placement & Tracking
Requirements Traceability	FR-4
Purpose	Facilitate order placement and real-time tracking.
Priority	High
Preconditions	The user must have items in the cart and a valid payment method.
Postconditions	The order is placed, and the user receives tracking updates.
Actors	Customer, System
Extends	N/A
Includes	UC5: Shipping & Tracking Updates
Main Success Scenario	1. The user confirms the order. 2. The system processes the payment. 3. The order is assigned to a courier. 4. Tracking details are sent to the user. 5. The user receives real-time tracking updates.
Alternate Flows	1. If payment fails, the system prompts the user to retry. 2. The user can cancel the order within a specified time frame.
Exceptions	1. Delays in courier service prevent accurate tracking updates. 2. The order tracking system experiences downtime.

3.7.5 Use Case Diagram:

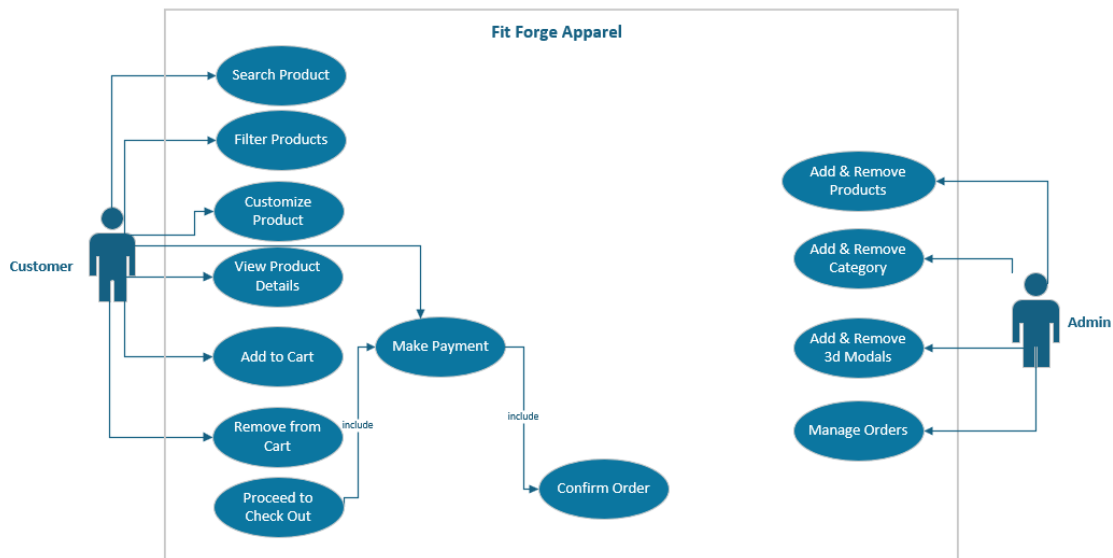


Figure 1 Use Case Diagram of Fit Forge Apparel System

3.8 Storyboards

3.8.1 Storyboard 1: Admin

Overview

The Admin oversees system operations, manages products, processes orders, and monitors customer interactions. The goal is to maintain smooth platform functionality while ensuring a seamless shopping experience for users.

Storyboard Flow

- Admin logs into the system using secure credentials.
- Upon successful login, the Admin Dashboard is displayed, showing system statistics, recent user activities, and pending orders.
- Admin navigates to Product Management, where they can add, edit, or delete products.
- Admin accesses Order Management to view pending orders, update statuses (Processing, Shipped, Delivered), and manage cancellations or refunds.
- Admin reviews Customer Feedback and inquiries, responding to issues or concerns raised by users.
- Admin checks System Logs & Security Settings to monitor system health and prevent unauthorized activities.

Activity Diagram:



Figure 2 Activity Diagram of Admin

3.8.2 Storyboard 2: User

Overview

Users interact with the system primarily for product customization, purchasing, and order tracking. The objective is to provide a seamless e-commerce experience where users can either purchase existing products or customize their own using the 3D customization tool, followed by a secure checkout process.

Storyboard Flow

- User visits the homepage and logs in or registers for a new account.
- After logging in, the user browses products through categories or search functionality.
- The user has two options:
- Purchase an existing product → Adds the selected product directly to the cart.
- Customize a product using the 3D Customization Tool → Modifies colors, adds logos, and personalizes the design.
- If customized, the user saves the design and adds the product to the cart.
- The user proceeds to checkout, enters shipping details, and completes payment securely.
- After order confirmation, the user receives real-time tracking updates via the Order Tracking Page.
- The user is notified when their order is out for delivery, and upon receiving the item, they can submit feedback and reviews.

Activity Diagram:

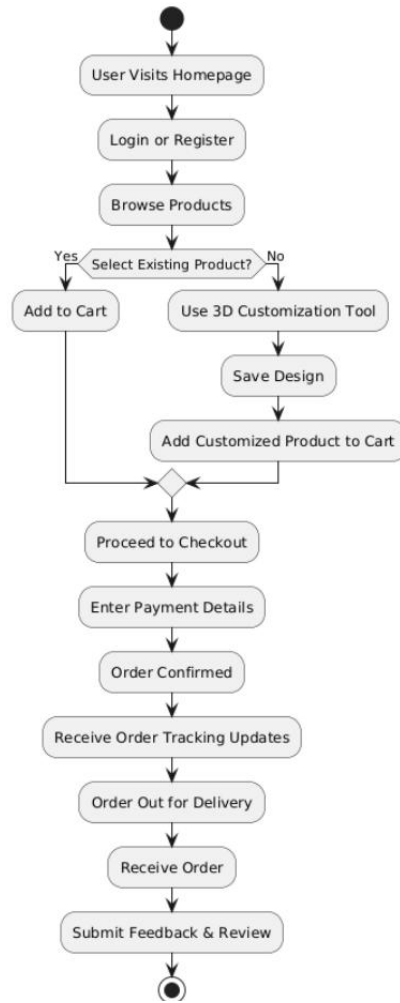


Figure 3: Activity Diagram of User

3.9 Summary

The process of requirements identification and use case analysis is essential for ensuring the Fit Forge Apparel system meets user expectations and business goals. This involves gathering, analyzing, and refining both functional requirements (e.g., product browsing, customization, order management, payment processing) and non-functional requirements (e.g., performance, security, usability, reliability). Use case analysis defines user interactions, such as customizing products, placing orders, and tracking shipments, ensuring alignment between system components and user needs. This structured approach minimizes design flaws, reduces development risks, and enhances user satisfaction, resulting in a high-performance, user-friendly e-commerce platform for custom sportswear.

Chapter No 4

Software Design Specification

4.1 System Design

4.1.1 Product Perspective

Fit Forge Apparel is an e-commerce platform that allows consumers to customize sportswear using an interactive 3D interface. The system runs as a web application, with a React interface for real-time modification, a Laravel backend for order processing and business logic, and a MySQL database for data storage. The platform is meant to be responsive and accessible on a variety of platforms, including PCs, tablets, and mobile phones.

The system interacts with various external services:

- **Payment Gateway (Payoneer & Visa API)**
Enables safe and secure transactions for purchases.
- **Email & Notification Service**
Sends out order confirmations, promotional emails, and updates.

To ensure that data is exchanged smoothly between the frontend, database, and external connections, the backend uses RESTful APIs.

4.1.2 Design Constraints

The Design Constraints of the System are as follow:

- **Performance Requirements**
 - The system should support real-time 3D customization with low latency.
 - Optimize database queries to handle high volume user interactions efficiently.
 - The website should load within 3 seconds for a seamless user experience.
- **Scalability**
 - The system should be able to handle multiple concurrent users customizing products without lag.
 - Cloud storage and content delivery networks (CDNs) will be employed to improve asset delivery.
- **Security & Compliance**
 - User authentication and session management should be implemented using JWT (JSON Web Tokens).
 - Secure payment processing must comply with PCI DSS standards.
 - Encrypting user data helps prevent unauthorized access.
- **Device Compatibility & Responsiveness**
 - The frontend must be designed using responsive frameworks to work across different screen sizes.
 - WebGL-powered 3D customization must be optimized for mobile and desktop performance.

4.2 Design Considerations

4.2.1 Assumptions and Dependencies

The design of Fit Forge Apparel is based on the following assumptions and dependencies:

- **Stable Internet Connection**
Users must have a stable internet connection for seamless 3D customization and order processing.
- **Browser Compatibility**
The application is assumed to work with modern browsers that support WebGL and JavaScript (Chrome, Firefox, Edge, and Safari).
- **Third-Party Services Availability**
The platform depends on Payoneer & Visa API for payments, and email/notification services. Any of services may impact operations if its down.
- **Device Performance**
Users require a decently capable device with GPU capabilities for smooth real-time 3D rendering.
- **Database Scalability**
To effectively handle increased traffic, the MySQL database structure assumes optimized queries and caching methods.

4.2.2 Limitations

While the system is built for maximum performance, the following constraints may affect functionality:

- **3D Customization Performance**
Complex models with high-resolution textures may cause slower loading times on low-end devices.
- **Mobile Compatibility for 3D Rendering**
Although the website is responsive, full 3D customization may not operate properly on older mobile devices owing to restricted GPU capability.
- **Limited Offline Functionality**
Because it is a web-based platform, personalization, order placement, and payment processing all depend on an active internet connection.
- **Payment Gateway Restrictions**
The site uses Payoneer & Visa API, a payment processor with potential regional restrictions.

4.3 Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) ensures that all system design aspects are consistent with the functional requirements specified in the Software Requirements Specification (SRS). This aids in ensuring that all requirements have been correctly implemented in the system's architecture.

Table 29 Requirements Traceability Matrix

Requirement ID	Scope	Requirement Description	Design Specification
R1	Product Customization	Users will be able to choose, personalize, and view clothing in real time through the system.	Component: 3D Customization Engine
R2	User Management	User authentication, profiles, and order history tracking will all be supported by the system.	Component: User Authentication
R3	Payment Processing	Users will be able to obtain transaction data and make safe payments through Payoneer & Visa API.	Component: Payment Gateway (Payoneer & Visa API)
R4	Order Management	Users will be able to place, monitor, and manage their orders using the system.	Component: Order Processing Module
R5	Inventory Management	The system will update availability, control stock levels, and alert users to stock changes.	Component: Inventory Database
R6	Shipping & Delivery	The technology will provide choices for order tracking and compute delivery charges.	Component: Shipping & Logistics API
R7	Performance Optimization	The system must load product previews in three to five seconds or less for seamless user experience.	Component: Performance Optimization (Caching & CDN)

4.4 Design Models

The design models offer a structural representation of the system architecture, emphasizing the interactions between various entities, data flow, and system behavior. The system is designed using the object-oriented development (OOD) approach, which ensures modularity, scalability, and maintenance.

4.4.1 Design Class Diagram (DCD)

The Design Class Diagram (DCD) shows the structure of the entire system, including domain classes, supporting classes, and their interactions. It is created with UML notation and ensures that object-oriented concepts like abstraction, polymorphism, inheritance, and encapsulation are used correctly.

Key Components of the DCD

- **Domain Classes**
Represent the system's main components, including the user, product, modal, order, cart, payment, and tracking.
- **Supporting Classes**
Controller classes that use the Model-View-Controller (MVC) design to make interactions easier.
- **Relationships**

Includes associations, aggregations, compositions, and generalizations.

Key Classes & Their Design

- **User Class:**
 - **Attributes:**
id, name, email, password,
 - **Methods:**
register(), login(), updateProfile(), placeOrder(), addToCart()
 - **Description:**
Represents a registered user with authentication and order management functions.
- **Product Class:**
 - **Attributes:**
id, name, description, image, price, status, discount, stock, category_id, supplier_id, admin_id
 - **Methods:**
addProduct(), updateProduct(), deleteProduct(), getProductDetails()
 - **Description:**
Stores information about products listed in the e-commerce platform.
- **Category Class:**
 - **Attributes:**
id, name, description, description, status
 - **Methods:**
addCategory(), updateCategory(), deleteCategory(), getCategoryDetails()
 - **Description:**
Represents product categories for better organization and navigation.
- **Cart Class:**
 - **Attributes:**
id, product_id, user_id
 - **Methods:**
addToCart(), removeFromCart(), clearCart()
 - **Description:**
Each user has a cart to manage selected items before purchase.
- **Cart_Info Class:**
 - **Attributes:**
id, cart_id, product_id, quantity
 - **Description:**
Tracks product quantity within the cart.
- **Order Class:**
 - **Attributes:**
id, user_id, admin_id, shipper_id, order_date, amount
 - **Methods:**
placeOrder(), cancelOrder(), getOrderDetails()
 - **Description:**
Manages purchase transactions.

- **OrderInfo Class:**
 - **Attributes:**
id, order_id, product_id, quantity, price
 - **Description:**
Stores details of each product in an order.
- **Payment Class:**
 - **Attributes:**
id, order_id, user_id, amount, payment_method
 - **Methods:**
processPayment()
 - **Description:**
Handles payment processing for orders.
- **Admin Class:**
 - **Attributes:**
id, name, email, password
 - **Methods:**
manageUsers(), manageProducts(), manageOrders()
 - **Description:**
Admin manages system functionalities.
- **Model Class:**
 - **Attributes:**
Id, name, admin_id, description, modal_type
 - **Methods:**
addModel(), updateModel()
 - **Description:**
Stores product customization models.
- **TrackingDetail Class:**
 - **Attributes:**
id, order_id, status, estimated_delivery
 - **Methods:**
updateTracking()
 - **Description:**
Provides real-time tracking of shipped orders.
- **Customize Class:**
 - **Attributes:**
id, user_id, model_id, type, color, logo
 - **Methods:**
applyCustomization()
 - **Description:**
Allows users to customize apparel before ordering
- **Review Class:**
 - **Attributes:**
id, user_id, product_id, rating, comment
 - **Methods:**
addReview(), updateReview(), deleteReview(), getProductReviews()

- **Description:**
This class stores customer feedback, including ratings and comments for products. It helps other users make informed decisions and allows admins to monitor product performance.

1. Class Diagram:

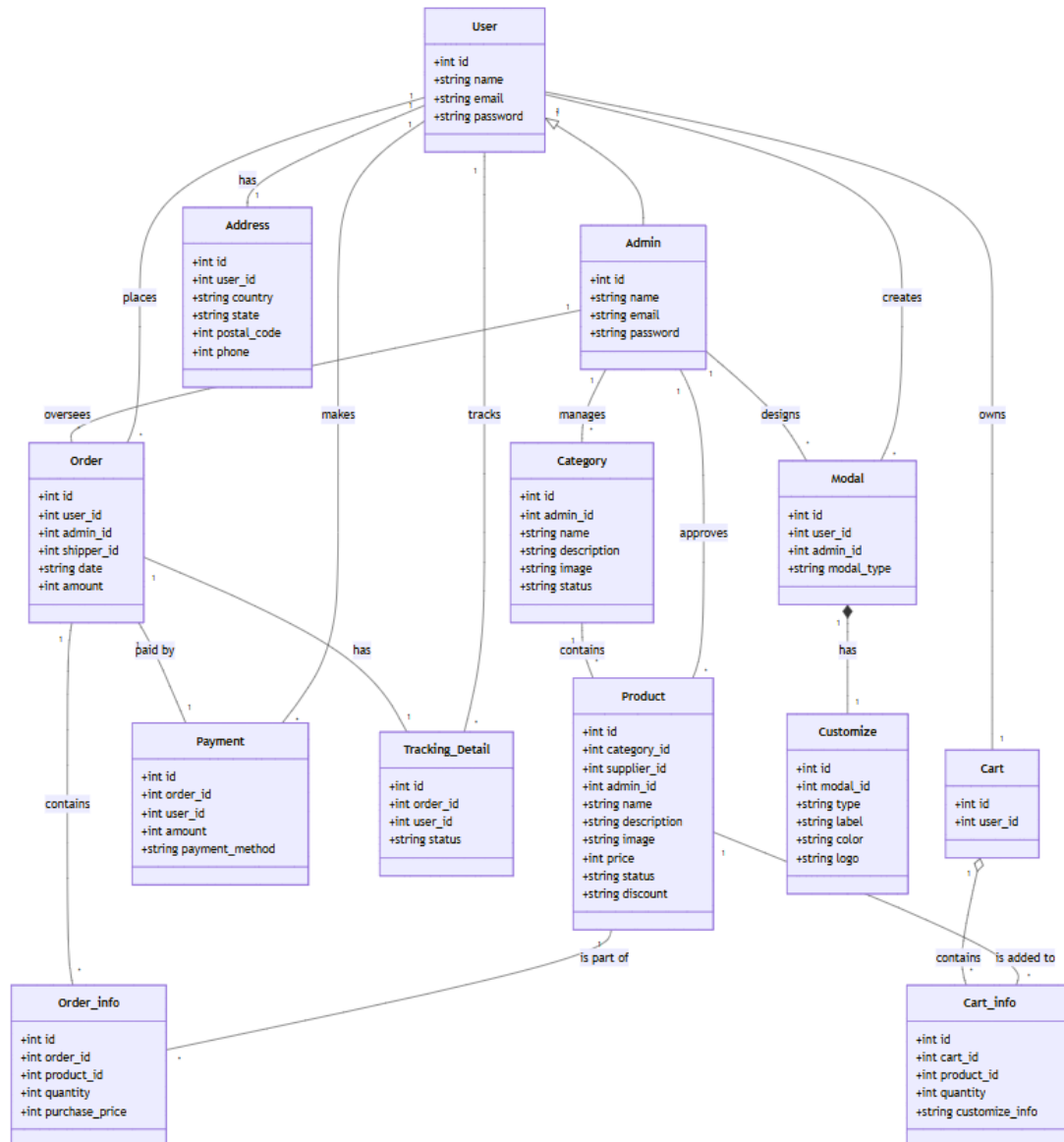


Figure 4: Class Diagram of Fit Forge Apparel

4.4.2 Interaction Diagram

The Sequence Diagram shows the interaction between the user, the system, and the database when placing an order with Fit Forge Apparel. It follows the UML standard format and is consistent with the class diagram shown earlier.

Actors & Components Involved:

- **User** The customer starts the order procedure.
- **Frontend (Website/App)**: The frontend (website/app) handles user input and routes requests to the backend.
- **Customization step added before order placement.**
- **Customize-Service verifies & applies customization before finalizing the order.**
- **Order-Controller**: Controls order placement logic.
- **Product-Service**: Verifies product availability and updates inventory.
- **Order-Service**: Creates and saves order details.
- **Payment-Service**: Processes and updates payment status.
- **Database**: Database stores order, user, and payment information.

Sequence of Events:

- The user selects customized or regular products and completes checkout.
- The Frontend makes a request to the Order-Controller to place the order.
- If the product is customized, the Order-Controller sends a request to Customize-Service to apply customization.
- The Customize-Service stores customization details and updates the product reference.
- The Order-Controller communicates with Product-Service to verify stock.
- If available, Order-Service generates the order.
- The Payment-Service processes the payment using the Payment Gateway.
- On successful payment, the order is confirmed, and the Database updates payment status.
- The system sends an order confirmation email to the user.

4.4.3 UML Sequence Diagram

1. Customization Modal:

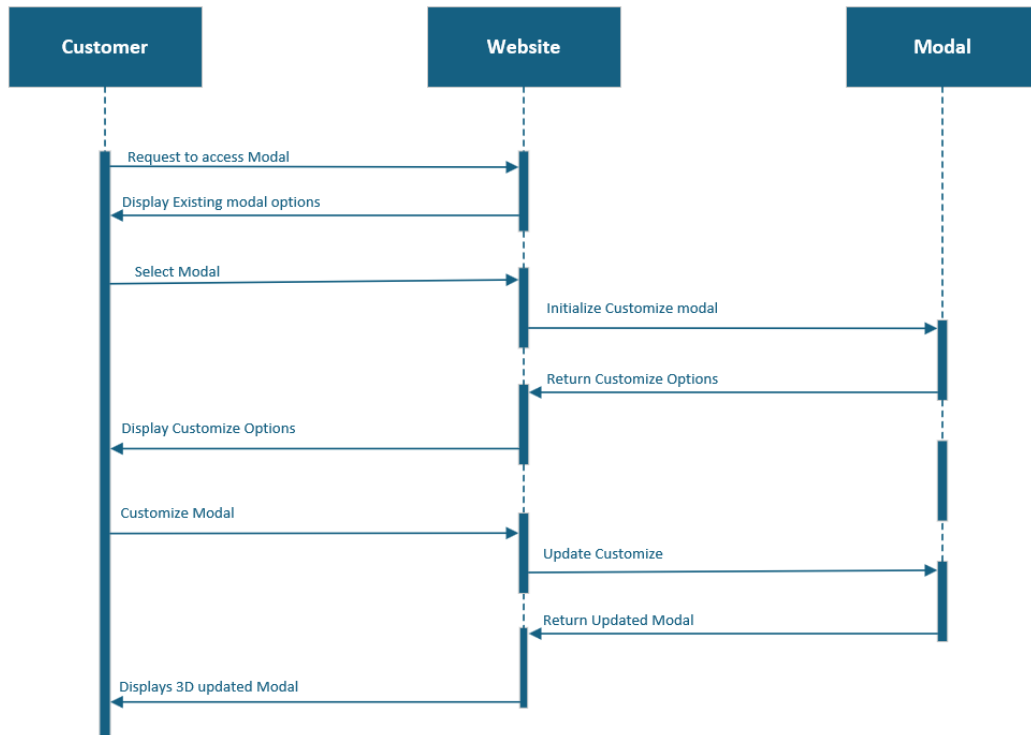


Figure 5: Sequence Diagram of Customize Modal

2. Save Modal:

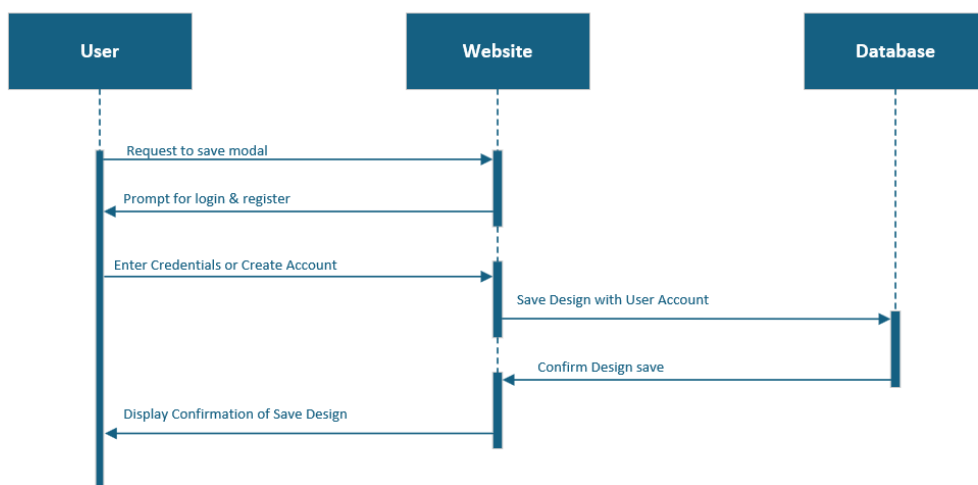


Figure 6: Sequence Diagram of Save Modal

3. Product Browsing Modal:

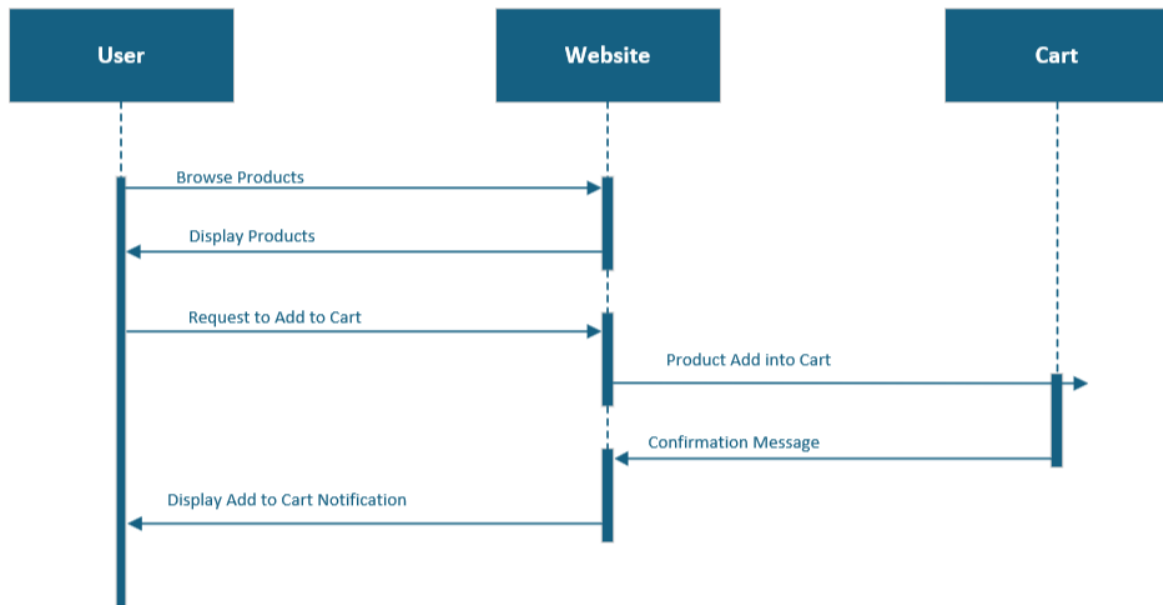


Figure 7: Sequence Diagram of Product Browsing Modal

4. Checkout Process:

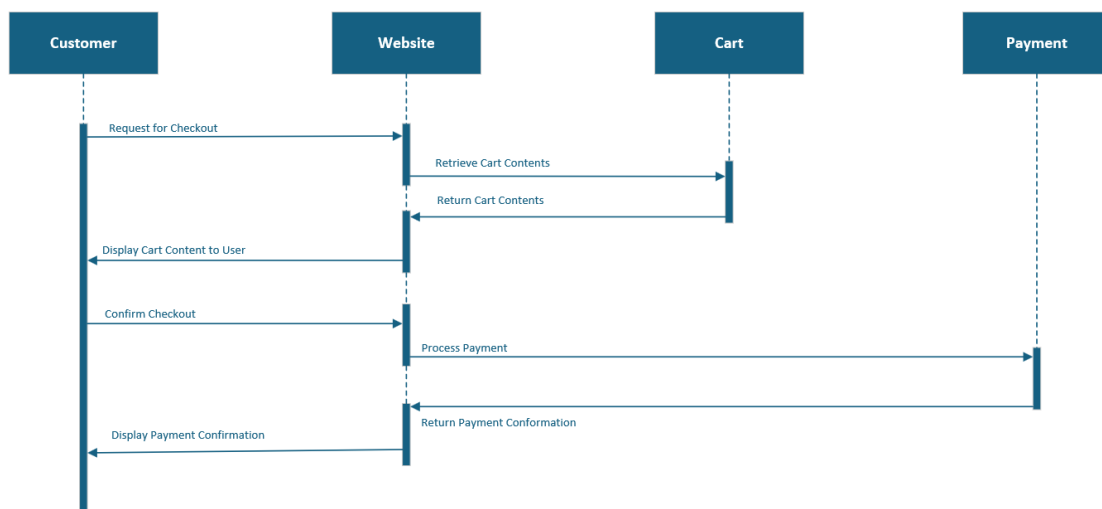


Figure 8: Sequence Diagram of Checkout Process

5. Login & Registration:

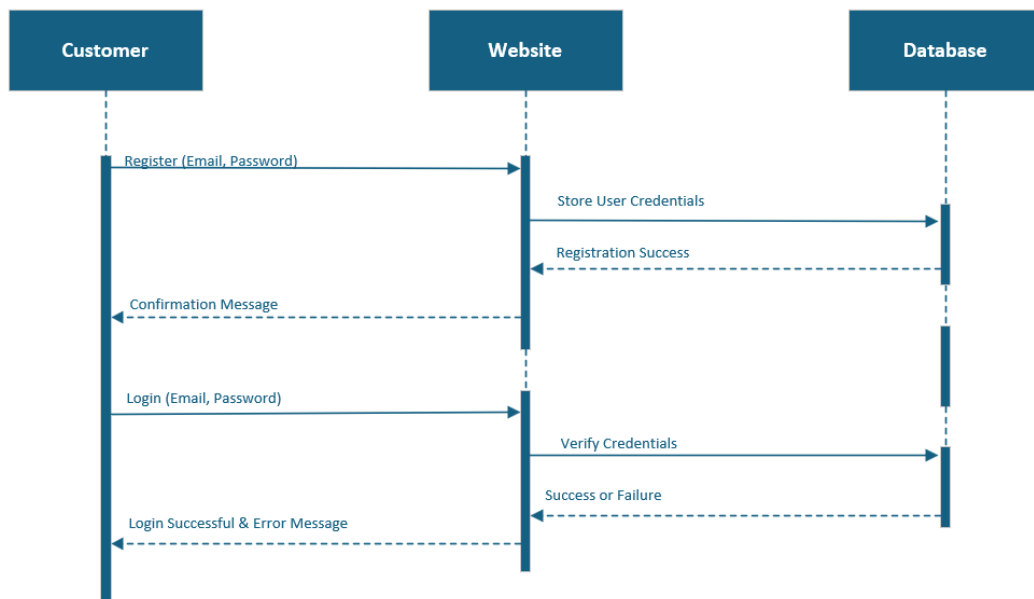


Figure 9: Sequence Diagram of Login & Registration

4.2.3 State Transition Diagram

A State Transition Diagram (STD) for Fit Forge Apparel now includes the customization process before order creation. It visualizes the order lifecycle, incorporating events like customization approval, payment success, shipment, and cancellation.

States in the Customization Process:

Table 30 : Phase 1 of Customization Process

Current State	Event/Trigger	Next State
1. Product Selected	User starts customization	Customizing
2. Customizing	User applies custom design (e.g., color, logo, text)	Customization Review
3. Customization Review	User confirms the final design	Customization Approved
4. Customization Approved	User proceeds to checkout	Order Created

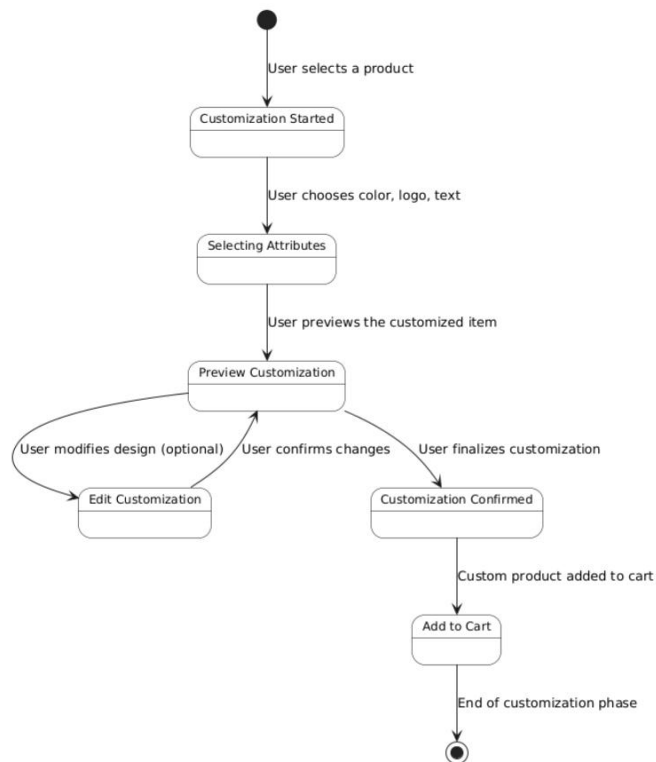


Figure 10: State Diagram of Customization Process

Table 31 Phase 2 of Order Processing Workflow (Common for All Orders)

Current State	Event/Trigger	Next State
Order Created	User places order	Payment Pending
Payment Pending	Payment Successful	Payment Confirmed
Payment Pending	Payment Failed	Order Cancelled
Payment Confirmed	Order processed by admin	Order Processing
Order Processing	Order dispatched by shipper	Shipped
Shipped	Order delivered to user	Delivered
Any State	User cancels order	Order Cancelled
Cancelled	Refund initiated (if applicable)	Refund Processed

State Transition Diagram for Order Processing & Delivery

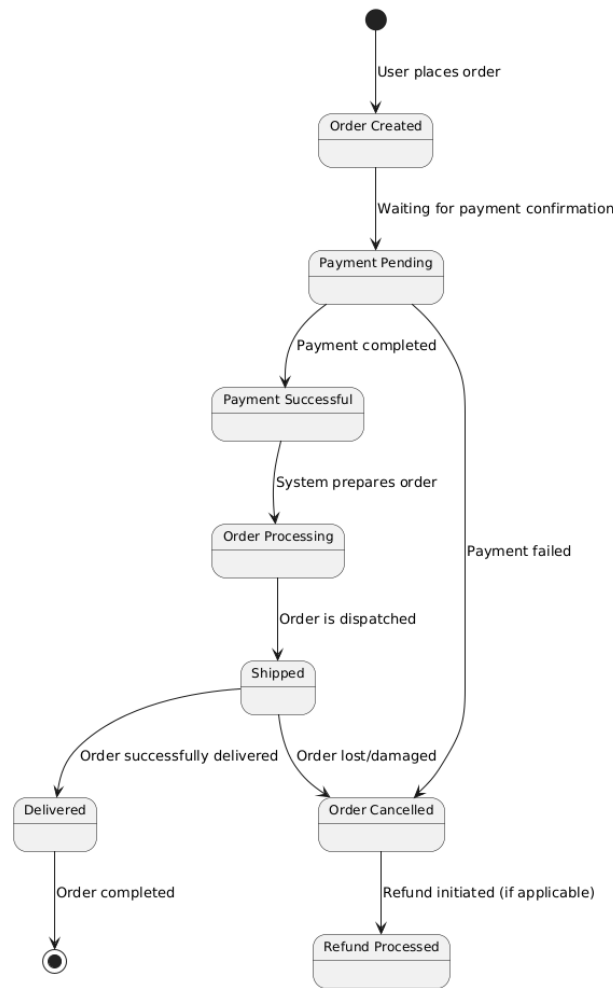


Figure 11: State Diagram of Order Processing Workflow

Diagram Representation (Description):

1. **Start State:**
 - If the user selects a pre-designed product, they move directly to order creation.
 - If the user customizes a product, they go through the customization phase first.
2. **Transitions:**
 - Arrows represent state changes triggered by user actions or system events.
3. **Final States:**
 - Delivered (Successful Purchase)
 - Refund Processed (Order Cancelled & Refunded)

4.5 Architectural Design

Fit Forge Apparel uses a multi-tiered architecture to ensure scalability, maintainability, and versatility. The system is built using the MVC (Model-View-Controller) paradigm, which ensures clear separation of concerns and effective handling of user interactions, business logic, and data administration.

System Components & Relationships

1. Frontend (User Interface) — View Layer

- Developed with React, Three.js, and Tailwind CSS for real-time 3D customization.
- Our interactive e-commerce platform allows consumers to design and visualize clothes before making a purchase.
- Uses RESTful APIs to communicate with backend systems.

2. Backend (Business Logic) — Controller Layer

- Developed with Laravel for business logic, order processing, and user authentication.
- Manages product customization logic, user profiles, and payment processes.
- Uses APIs to connect with external services like payment gateways and shipping companies.

3. Database (Data Management) — Model Layer

- MySQL stores product details, user data, orders, and payment transactions.
- Utilizes a normalized schema to improve efficiency and reduce redundancy.
- Encryption and access control ensure consistent and secure data.

4. External Services & Integrations

- Use a secure payment gateway, such as Payoneer & Visa , for transactions.
- Shipping API for tracking and managing deliveries.
- Authentication service for OAuth logins (e.g. Google, Facebook).

Diagram Representation

The Architectural Diagram includes:

- The User Interface (React, Three.js), which handles customer interactions and customizations.
- The API Gateway routes requests between the frontend and backend.
- Controller (Laravel) that handles authentication, business logic, and API requests.
- The database (MySQL) stores and retrieves system data.
- External APIs facilitate payment, delivery, and third-party authentication.

4.5.1 UML Component diagram

The UML Component Diagram depicts the high-level structure of the Fit Forge Apparel system, including the primary components and their relationships. The system employs a multi-tiered architecture based on the MVC (Model-View-Controller) pattern, which ensures modularity, scalability, and maintenance.

Key Components & Their Relationships

1. User Interface (Frontend) [View Layer]

- **Technologies:** React, Three.js, Tailwind CSS
- **Functionality:** Provides an interactive e-commerce experience with 3D model customization.
- **Interaction:** Sends user inputs and product configurations to the backend via RESTful APIs.

2. API Gateway (Middleware)

- **Functionality:** Routes requests between the frontend and backend services.

- **Interaction:** Ensures secure and efficient data transfer through Laravel API endpoints.
- 3. Backend (Business Logic) [Controller Layer]**
- **Technology:** Laravel
 - **Functionality:** Handles user authentication, order processing, and business logic for custom apparel design.
 - **Interaction:** Retrieves and updates data from the database and interacts with external APIs for payments and shipping.
- 4. Database (Data Management) [Model Layer]**
- **Technology:** MySQL
 - **Functionality:** Stores user profiles, product details, orders, and transaction history.
 - **Interaction:** Provides data access for the backend services.
- 5. External Services**
- **Payment Gateway:** Secures online transactions.
 - **Shipping API:** Tracks order deliveries.
 - **OAuth Authentication:** Supports login via Google, Facebook, etc.

Table 32: Component Mapping to Architecture Style (MVC Multi-Tiered)

Architecture Layer	Components	Description
View (Frontend)	React, Three.js, Tailwind CSS	Provides a user-friendly 3D customization interface
Controller (Backend)	Laravel API Gateway	Handles business logic and processes API requests
Model (Database)	MySQL Database	Stores user, order, and product data
External Services	Payment Gateway, Shipping API, OAuth Authentication	Ensures secure payments, deliveries, and logins

Diagram Representation

- Users use the frontend (React UI) to take and customize products.
- The frontend connects with the Laravel API (Controller Layer) to handle requests.
- The API Gateway manages request routing and connects with the MySQL Database (Model Layer).
- External APIs connect to the backend for payment processing, shipping, and authentication.

The UML diagrams are as followed:

- **Block Diagram**

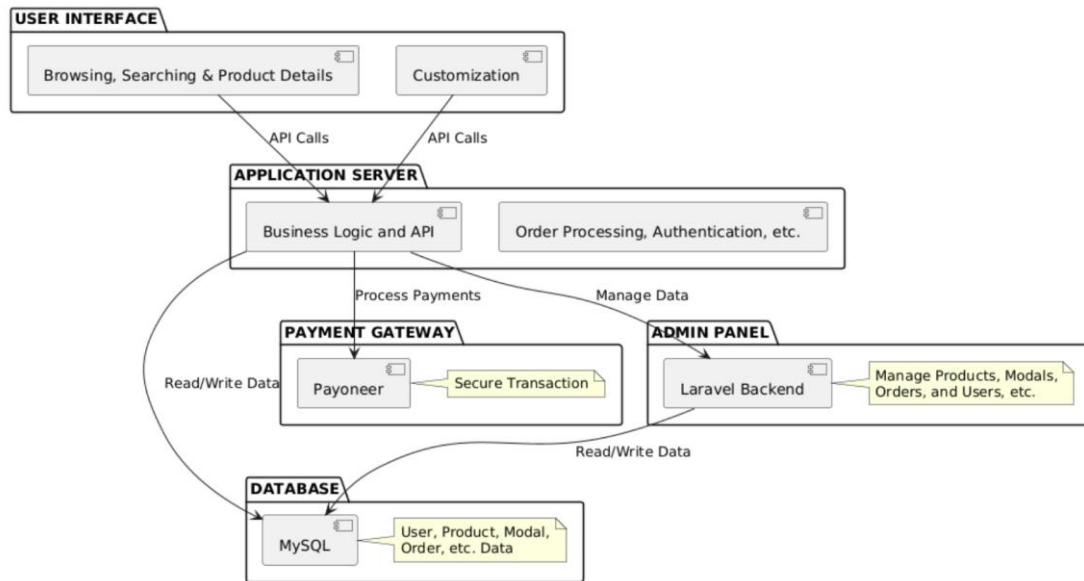


Figure 12 Block Diagram

- **Layer Diagram**

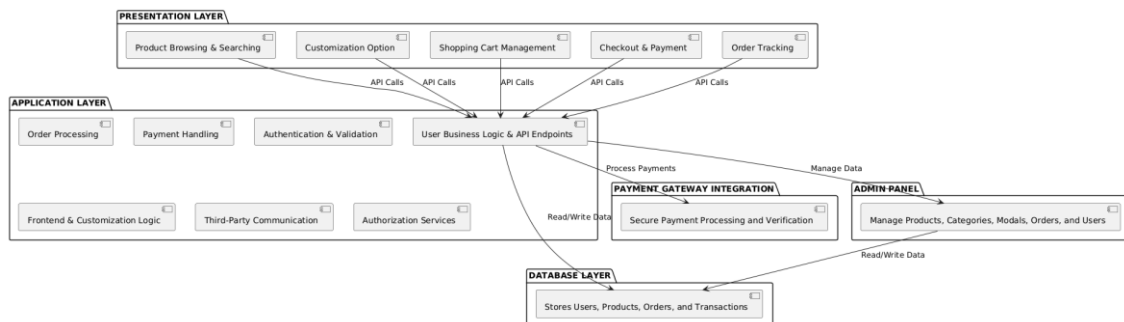


Figure 13: Layer Diagram

- **Deployment Diagram**

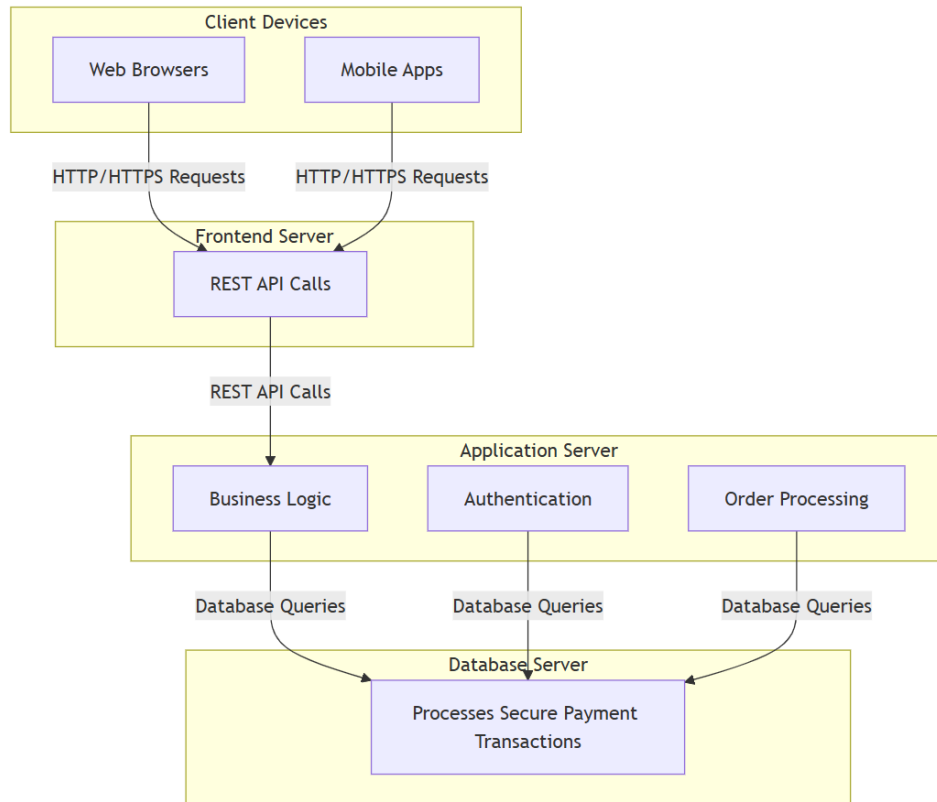


Figure 14: Deployment Diagram

- **ER Diagram**

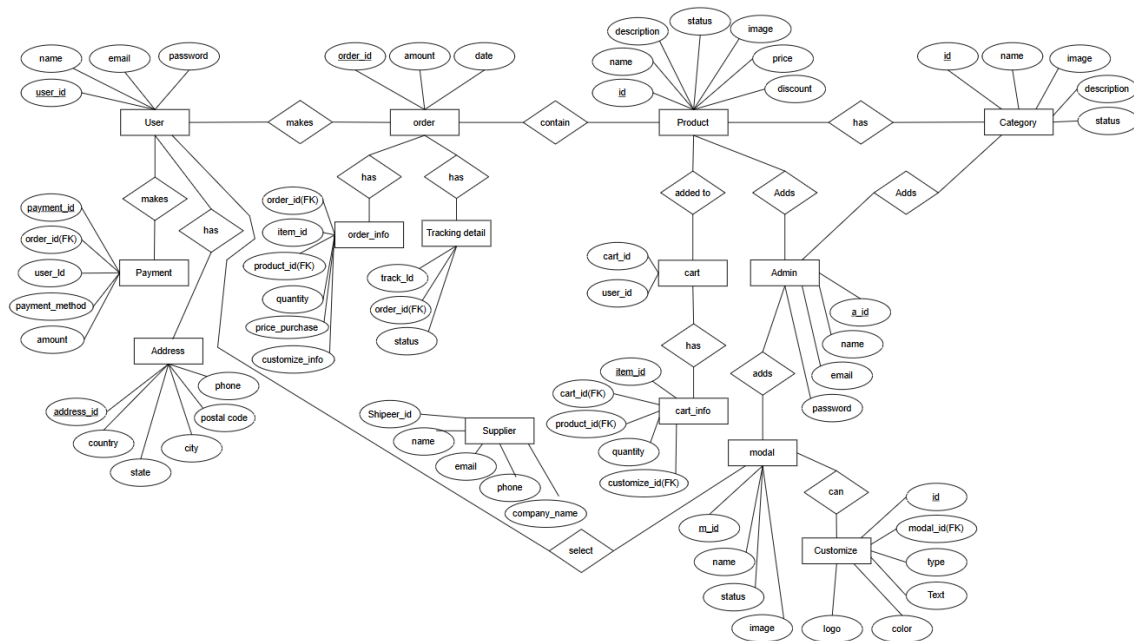


Figure 15: ER-Diagram

4.6 Data Design

The Fit Forge Apparel system is designed to manage user interactions, product customization, order processing, and payment processes efficiently. The data design ensures that information is appropriately structured, stored, and accessed, resulting in smooth system performance.

Data Storage & Processing:

- The system uses a relational database (MySQL) to efficiently store data.
- Normalized tables are used to arrange data and reduce redundancies.
- Optimized queries for excellent performance and quick retrieval.
- Ensure data integrity using secure CRUD activities (Create, Read, Update, and Delete).

4.6.1 Data Dictionary

- **Purpose:**
 - To provide a comprehensive and organized list of all system entities, attributes, and significant data elements.
 - Each entry includes the data type and a brief description of its purpose within the system.
- **Format:**
 - Presented in a two-column format:
 - **First Column:** Contains the terminology (e.g., object name, attribute, method, or method parameter).
 - **Second Column:** Provides the description, including data types, roles, or any other relevant information.

Structured Approach: Functions and Function Parameters

- **Purpose:**
 - For systems following a structured programming paradigm, this section lists all functions along with their parameters.
 - Each function is described in terms of its functionality, and each parameter is specified with its data type and role within the function.
- **Format:**
 - **Function Name:** Description of the function's purpose.
 - **Parameters:** List of parameters with their data types and roles.

Object-Oriented (OO) Approach: Objects, Attributes, Methods, & Method Parameters

- **Purpose:**
 - For systems using the Object-Oriented (OO) approach, this section lists:
 - Objects and their attributes, along with a description.
 - Methods associated with each object, including their method parameters.
- **Format:**
 - **Object Name:** Description of the object's purpose.
 - **Attributes:** List of attributes with their data types and descriptions.

- **Methods:** List of methods with their parameters (data types and roles) and a description of their functionality.

4.7 User Interface Design

The User Interface Design of the Customize E-Commerce Website are as follow :

4.7.1 Screen Images

- Main page

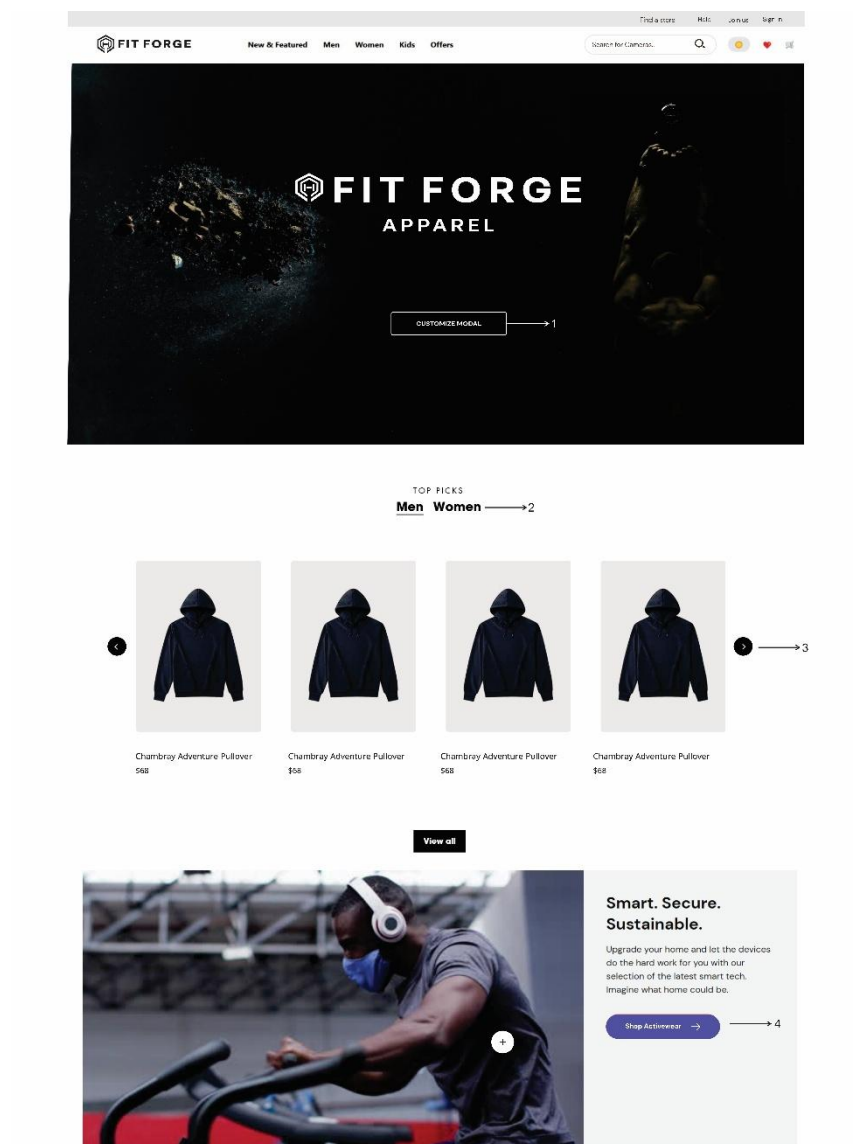


Figure 16: Main page

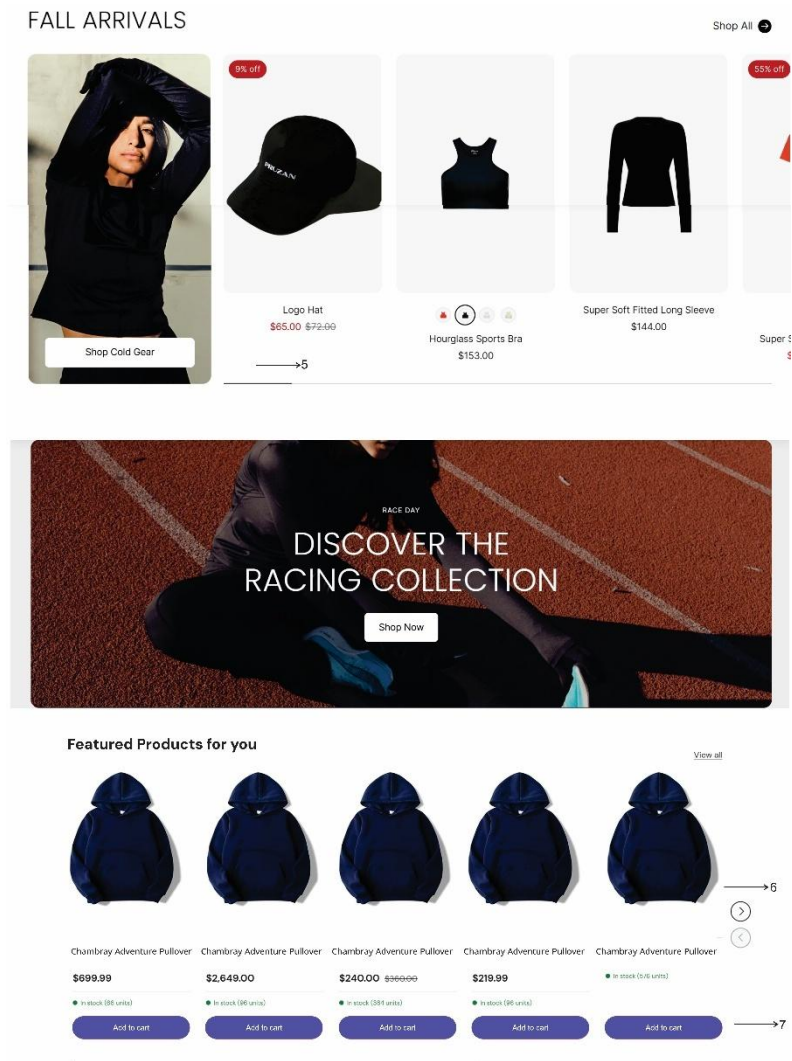
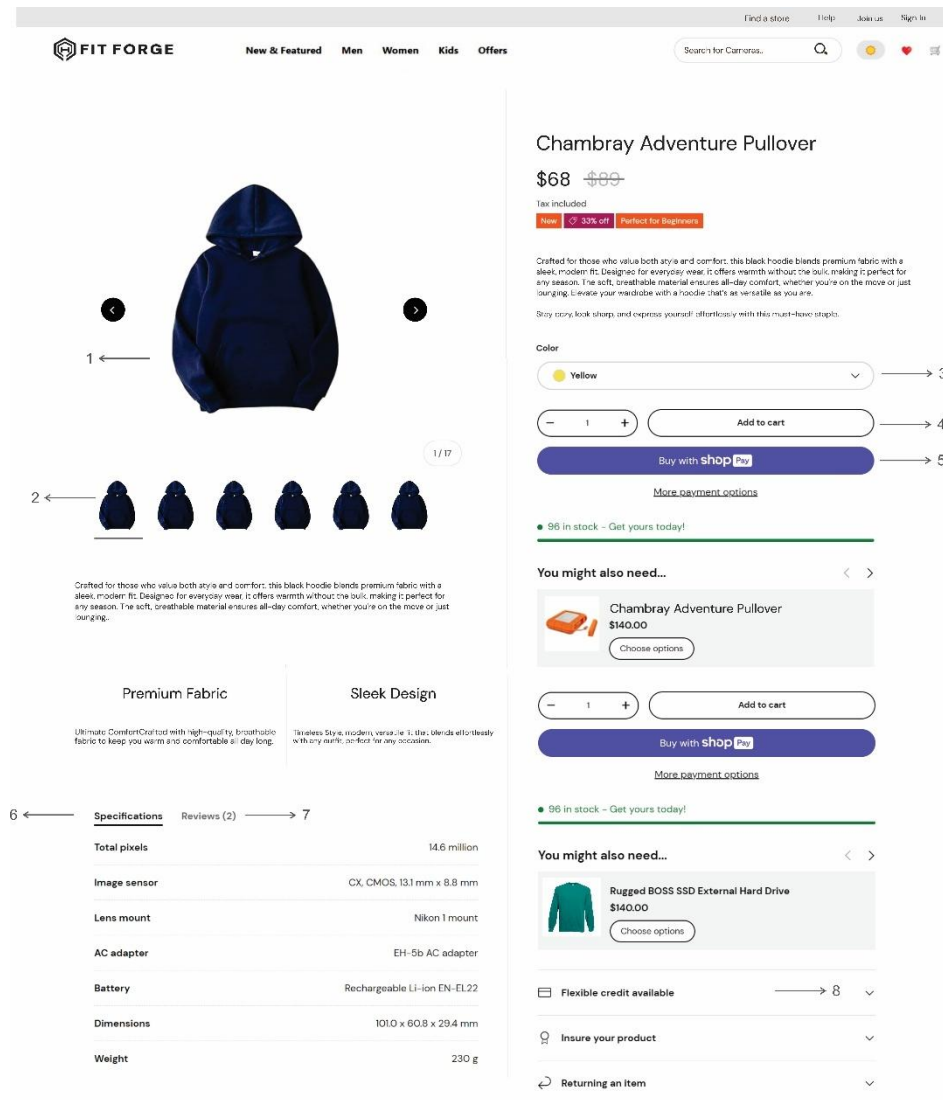


Table 33 : Use case of Main Page

Step No.	Action
1	The user clicks the Customize Model button to personalize a product.
2	The user selects between Men or Women categories in the Top Picks section.
3	The user browses through the featured products and clicks on a product for more details.
4	The user clicks on Shop Activewear to explore more collections.
5	The user scrolls through Fall Arrivals and selects a product of interest.
6	The user explores the Featured Products section and views different pricing options.
7	The user clicks the Add to Cart button to purchase a selected product.
8	The user clicks Shop Cold Gear to explore winter apparel.
9	The user scrolls through the Featured Products section, reviewing various pricing options.
10	The user clicks the Add to Cart button to add a selected item for purchase.

- **Product Select Page**



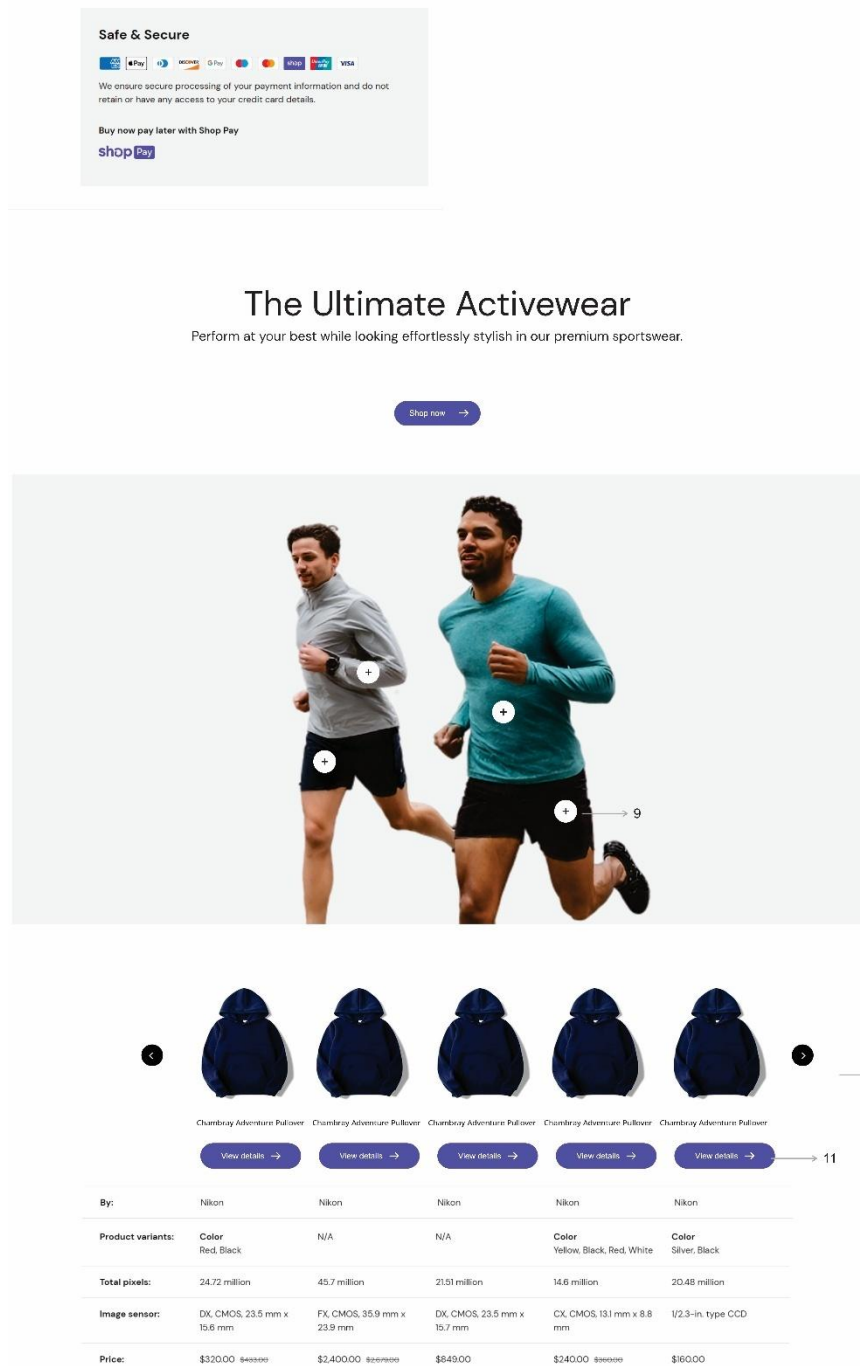


Figure 17: Product Select Page

Table 34: Use Case of Product Select Page

Step No.	Action
1	The user views the main product image
2	The user clicks the image thumbnails to explore different product angles.
3	The user selects the product color option from the dropdown.
4	The user adjusts the quantity of the product before purchase.
5	The user clicks Add to Cart to add the item to their shopping cart.

6	The user navigates to the Specifications tab for product details.
7	The user switches to the Reviews section to check customer feedback.
8	The user explores Flexible Credit and other payment options.
9	The user interacts with the hotspots on the image to explore product details.
10	The user navigates through the product slider to check different items.
11	The user clicks View Details on a product to open its detailed page.

• Product Search Page

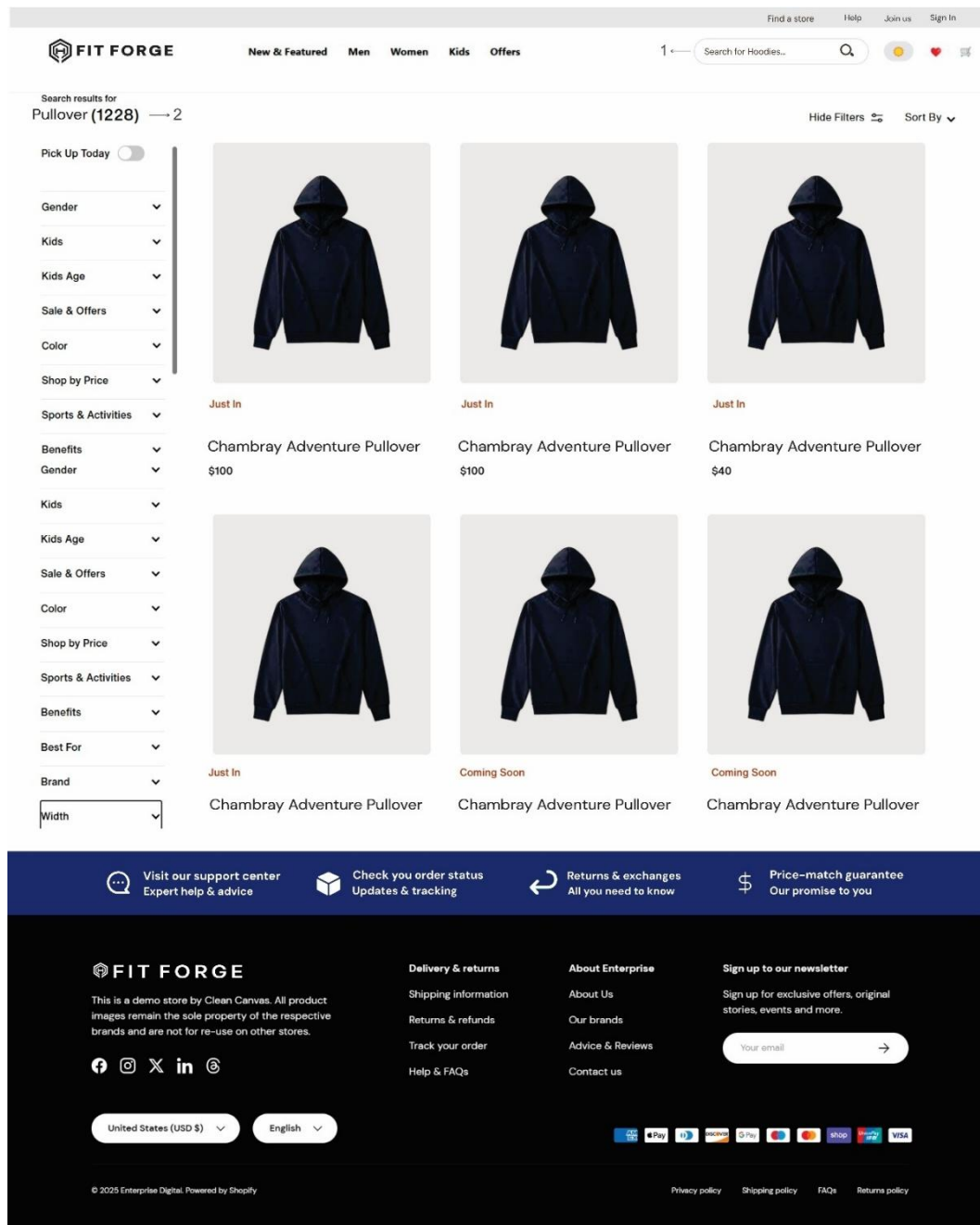


Figure 18: Product Search Page

Table 35 : Use Case of Product Search & Filtering

Step No.	Action
1	The user enters a search query (e.g., Hoodies) in the Search Bar.
2	The system displays search results with the total number of matching products.
3	The user can apply filters from the left sidebar (e.g., Gender, Kids Age, Sale & Offers, Color, Price, Brand).
4	The user can toggle the Pick Up Today option to see available products for pickup.
5	The system updates and refines the displayed products based on the applied filters.
6	The user can Sort By different criteria (e.g., price, new arrivals, best sellers).
7	The user selects a product to view its details by clicking on it.

- **Login Page**

Figure 19: Login Page

Table 36 : Use Case of User Login

Step No.	Action
1	The user enters their Email Address in the input field.
2	The user enters their Password in the input field.
3	The user clicks the "Sign In" button to log in.
4	If the user forgets their password, they can click on "Forgot Your Password?" to reset it.
5	If the user does not have an account, they can click on "Create an Account" to register.

- Registration Page

The screenshot shows the registration page for FIT FORGE. It is divided into two main sections: 'PERSONAL INFORMATION' and 'SIGN-IN INFORMATION'. The 'PERSONAL INFORMATION' section includes fields for 'First Name' (step 1), 'Last Name' (step 2), a checkbox for 'Sign Up for Newsletter' (step 3), and a checkbox for 'Allow remote shopping assistance' (step 4). The 'SIGN-IN INFORMATION' section includes fields for 'Email' (step 5), 'Password' (step 6), and 'Confirm Password' (step 7). Below these fields is a checkbox for agreeing to terms and conditions (step 8). At the bottom, there are two buttons: 'Back' (step 9) and 'Create an Account' (step 8).

Figure 20: Registration Page

Table 37 : Use Case of User Registration

Step No.	Action
1	The user enters their First Name in the input field.
2	The user enters their Last Name in the input field.
3	The user checks the Sign Up for Newsletter option (optional).
4	The user checks the Allow Remote Shopping Assistance option (optional).
5	The user enters their Email Address in the input field.
6	The user enters their Password in the input field.
7	The user re-enters their Password to confirm.
8	The user agrees to the terms by checking the consent checkbox.
9	The user clicks the Create an Account button to register.
10	If the user wants to exit, they click the Back button to return.

- Cart

The screenshot shows the cart page for FIT FORGE. It features a 'Your cart' section with a message 'Your order is eligible for free shipping!'. Below this, there are two items in the cart: a 'Nikon COOLPIX A1000 Point & Shoot Camera' (step 1) and a 'Nikon 1 S2 Point & Shoot Camera' (step 4). To the right, there is an 'Order summary' section (step 3) showing a subtotal of \$480.00 USD. Below the subtotal, there are buttons for 'Checkout' (step 3), 'shop Pay' (step 4), and 'Pay' (step 5). At the bottom, there is a 'Continue shopping' link (step 6). Below the cart items, there is a 'You may also like...' section with two recommended products: 'Rugged BOSS SSD External Hard Drive' (step 7) and 'ecobee SmartSensor 2-pack' (step 7).

Figure 21 : Cart Page

Table 38 : Use Case of Shopping Cart & Checkout

Step No.	Action
1	The user views the Shopping Cart page with all selected products.
2	The user can increase or decrease the quantity of each item.
3	The user clicks the Checkout button to proceed with payment.
4	The user can enter an Order Note for special instructions.
5	The user selects a Payment Method (e.g., Shop Pay, Google Pay, etc.).
6	The user clicks Continue Shopping to return to the store.
7	The system suggests related products under You may also like... for additional purchases.
8	The system displays a summary of the order total with applied taxes and discounts.
9	The system confirms the checkout and redirects the user to the payment gateway.

- **Modal Page**

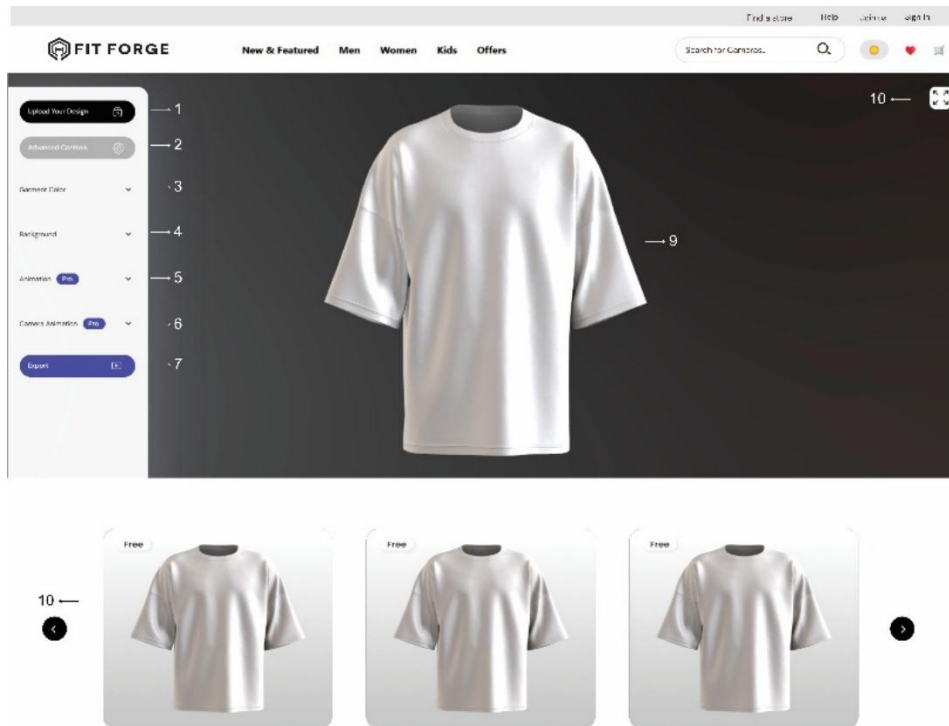


Figure 22: Modal Page

Table 39 : Use Case of 3D Product Customization

Step No.	Action
1	The user clicks on Upload Your Design to add a custom logo or artwork.
2	The user selects Garment Color to change the base color of the apparel.
3	The user chooses a Background Color for preview customization.
4	The user adjusts the Zoom level to inspect the design details.
5	The user enables or disables the Camera Auto Rotate option for better viewing angles.

6	The user clicks Export to save or share the customized design.
7	The system renders and updates the 3D Preview in real time.
8	The user can select different apparel variations from the available options.
9	The system displays a flat garment template with applied customization.
10	The user can proceed to purchase the customized product or save it for later.

4.8 Design Decisions

Fit Forge Apparel was designed with critical architectural and algorithmic decisions to provide scalability, performance, security, and maintainability. The following are the primary design decisions and their justifications:

1. Architectural Pattern – MVC (Model-View-Controller)

- The system uses the MVC design to effectively isolate issues.
- The model uses Eloquent ORM from Laravel to manage database interactions.
- The view is designed using React and Three.js for real-time 3D product modification.
- The controller oversees business logic and system workflows.
- Justification: improves maintainability, modularity, and simplifies debugging.

2. Object-Oriented Programming (OOP)

- The backend is structured using OOP principles (Encapsulation, Abstraction, Inheritance, Polymorphism).
- Laravel framework is used to define classes for Users, Products, Orders, and Cart with reusable methods.
- Justification: Promotes code reusability, scalability, and security.

1. Database Normalization (3NF – Third Normal Form)

- The database is created with 3NF to reduce redundancy and enhance query performance.
- Tables are efficiently structured with foreign keys to ensure referential integrity.
- Justification: Reduces data duplication, increases consistency, and accelerates query execution.

2. Algorithmic Choices

- The recommendation system uses a collaborative filtering algorithm to offer products based on user behavior.
- Improve search speed with full-text search and indexing approaches.
- Justification: Improves user experience through personalized recommendations and faster searches.

3. Security Measures

- Secure password storing with Bcrypt hashing algorithm.
- Input validation and sanitization prevent SQL injection and XSS attacks.
- JWT Authentication provides secure API access for user authentication.
- Justification: Protects user data and prevents cyber threats.

4. API & External Integrations

- The system communicates between the React frontend and Laravel backend via RESTful APIs.

- Payment Gateway Integration: Secure transactions with Payoneer & Visa API.
- Justification: Ensures a safe payment process and supports future API integrations.

4.9 Summary

This chapter outlined the key design concepts and decisions for the Fit Forge Apparel system, focusing on scalability, security, efficiency, and user experience. The MVC architecture ensured modularity and maintainability, while OOP principles enhanced code reuse and security. Database normalization (3NF) improved data integrity and query performance. Secure authentication (JWT, Bcrypt) and RESTful APIs enabled seamless communication, and algorithms like collaborative filtering and full-text search enhanced user experience. These design choices ensure a robust, scalable, and future-ready e-commerce platform, highlighting the project's technical feasibility and long-term viability.

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